

Milestone 4: Network Design and Proposal

Group 3

Lewis University - Computer Infrastructure Capstone (CPSC 49300)

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1. Introduction

1.1 Purpose

This document presents the initial design and proposal for the computer infrastructure project supporting the tutoring centers in Romeoville, Chicago, and Aurora, IL. The goal is to establish a secure, efficient, and scalable network infrastructure that meets the operational needs of the tutoring centers while ensuring compliance with security standards and best practices.

1.2 Scope of this Proposal

- Network topology and equipment selection
- Internet Service Provider (ISP) recommendations
- Security implementation (firewalls, VPN, VLAN segmentation)
- Budget considerations and cost estimates
- Ongoing IT support strategies

2. Background and Theory

2.1 Introduction to Networking

- Local Area Network (LAN) – A network that connects computers and devices within a small geographical area, such as an office or home.
- Metropolitan Area Network (MAN) – A network covering a larger area than a LAN, typically spanning a city or campus.
- Wide Area Network (WAN) – A network that connects multiple LANs over long distances, often using public or leased communication lines.
- Internet – A global network of interconnected computers and devices that use standardized protocols for communication.
- LAN Technologies – Technologies such as Ethernet, Wi-Fi, and fiber optics used to create and maintain local networks.
- WAN Technologies – Communication methods such as MPLS, SD-WAN, and leased lines that enable long-distance networking.
- Wireless Technologies – Communication methods, including Wi-Fi, LTE, and Bluetooth, that allow devices to connect without physical cables.
- Security Mechanisms – Defensive strategies like firewalls, encryption, intrusion detection/prevention systems (IDS/IPS), and VLAN segmentation used to protect networks.
- VPN Technology – A method that securely connects remote users or sites using encrypted tunnels over the internet.
- Two industry standard conceptual frameworks used to understand and standardize network communication- OSI and TCP/IP.

OSI Model

The OSI (Open Systems Interconnection) Model is a framework that standardizes how different networking systems communicate. It is divided into seven layers, each with a specific function that helps facilitate seamless communication across different network devices and protocols.

Physical Layer (Layer 1)

Function: This layer deals with the actual hardware and transmission of raw data bits over a physical medium such as cables, radio waves, or fiber optics.

Key Responsibilities:

- Defines electrical, mechanical, and procedural characteristics of communication.
- Determines how bits are represented (e.g., voltage levels, modulation).
- Handles physical connections (e.g., Ethernet cables, wireless signals).

Examples:

- Cables: Ethernet, Fiber Optic, Coaxial.
- Devices: Hubs, Repeaters, Network Interface Cards (NIC).
- Wireless Transmission: Bluetooth, Wi-Fi signals.

Data Link Layer (Layer 2)

Function: This layer is responsible for node-to-node data transfer, as well as error detection and correction.

Key Responsibilities:

- Divides data into frames and handles MAC (Media Access Control) addressing.
- Uses switches and bridges for efficient data forwarding.
- Error detection using Cyclic Redundancy Check (CRC).

Examples:

- Protocols: Ethernet, Wi-Fi (IEEE 802.11), MAC addresses.
- Devices: Switches, Bridges.

Network Layer (Layer 3)

Function: Manages routing and ensures data packets reach the correct destination across different networks.

- Uses logical addressing (IP addresses) to identify source and destination.
- Determines the best route for data using routing algorithms.
- Handles packet forwarding, fragmentation, and reassembly.

Examples:

- Protocols: Internet Protocol (IP), ICMP (ping), RIP, OSPF, BGP.
- Devices: Routers, Layer 3 switches.

Transport Layer (Layer 4)

Function: Ensures end-to-end communication, error handling, and reliability.

- Segmentation and reassembly of data into smaller packets.
- Flow control: Manages data transmission speed to prevent congestion.
- Error detection and correction via acknowledgments and retransmissions.

Examples:

- Protocols: TCP (Transmission Control Protocol), UDP (User Datagram Protocol).
- Usage: TCP ensures reliable communication (e.g., web browsing), while UDP is faster but less reliable (e.g., video streaming, VoIP).

Session Layer (Layer 5)

Function: Establishes, maintains, and terminates communication sessions between applications.

- Manages session creation, maintenance, and closure.
- Enables devices to resume interrupted sessions.
- Facilitates authentication and authorization.

Examples:

- Protocols: NetBIOS, RPC (Remote Procedure Call), PPTP.
- Usage: Online banking, file transfers, video conferencing.

Presentation Layer (Layer 6)

Function: Translates, encrypts, and compresses data so different applications can understand it.

- Data translation: Converts data formats between sender and receiver.
- Encryption & Decryption: Ensures secure data transfer.
- Compression: Reduces file size for efficient transmission.

Examples:

- Protocols: SSL/TLS (Secure Sockets Layer), JPEG, MP3, ASCII.
- Usage: Secure web browsing (HTTPS), multimedia streaming.

Application Layer (Layer 7)

Function: The topmost layer where users interact with applications over a network.

- Provides network services to end-users (email, file transfer, web browsing).
- Handles protocols that define how applications send/receive data.

Examples:

- Protocols: HTTP/HTTPS (web browsing), FTP (file transfer), SMTP (email).
- Usage: Google Chrome (HTTP), Outlook (SMTP), Zoom (VoIP).

TCP/IP Model

Link Layer (Network Access Layer)

Function: Responsible for physical transmission of data between devices within the same network.

- Defines how data packets are transmitted over a physical network.
- Handles MAC addressing, error detection, and access control.
- Enables communication within the same network (LAN/Wi-Fi).

Protocols & Technologies:

- Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11)
- ARP (Address Resolution Protocol)
- PPP (Point-to-Point Protocol)

Devices Used: Switches, Hubs, Network Interface Cards (NICs), Modems.

Internet Layer (Network Layer)

Function: Handles packet forwarding, addressing, and routing between different networks.

- Assigns IP addresses and routes data between networks.
- Determines the best path for data to reach its destination.
- Fragmentation and reassembly of large packets.

Protocols & Technologies:

- IP (Internet Protocol) – IPv4 & IPv6
- ICMP (Internet Control Message Protocol) – Used for diagnostics (ping).
- IGMP (Internet Group Management Protocol) – Used for multicast traffic.
- Routing Protocols: OSPF, BGP, RIP.

Devices Used: Routers, Layer 3 Switches.

Transport Layer

Function: Ensures reliable or fast data transmission between devices.

- Segmentation and reassembly of data into packets.
- Manages error detection, retransmission, and flow control.
- Provides different levels of reliability and speed based on the protocol used.

Protocols:

- TCP (Transmission Control Protocol) – Reliable, connection-oriented (e.g., Web Browsing, Email).
- UDP (User Datagram Protocol) – Fast, connectionless (e.g., VoIP, Streaming, Gaming).

Usage:

- TCP ensures error-checking and ordered data delivery (e.g., HTTPS, Email).
- UDP is faster but does not guarantee delivery (e.g., Zoom, DNS queries).

Application Layer

Function: Direct interaction between user applications and the network.

- Defines how applications communicate over the internet.
- Uses protocols that enable email, file transfer, web browsing, etc..

Protocols & Services:

- Web Browsing: HTTP, HTTPS
- Email Services: SMTP (Sending), POP3/IMAP (Receiving)
- File Transfer: FTP, SFTP
- Remote Access: SSH, Telnet
- Domain Name Resolution: DNS (Domain Name System)

Examples:

- Google Chrome (HTTP/HTTPS)
- Outlook (SMTP, IMAP)
- Filezilla (FTP)
- Zoom & Skype (VoIP – SIP/RTP)

(TechTarget, n.d.)

2.2 Network Security Considerations- Common security threats, defensive mechanisms, and compliance requirements.

Common Security Threats

Distributed Denial-of-Service (DDoS) Attacks- Overwhelms a network, server, or website with excessive traffic, causing service disruptions. Will slow down or crash a system, making them unavailable to users. Firewalls and intrusion prevention systems (IPS) are used to mitigate DDOS attacks and detect abnormal traffic patterns.

Phishing Attacks- Cybercriminals trick users into revealing credentials, banking details, or sensitive data via deceptive emails or websites. This can compromise user credentials, financial loss, and unauthorized access to networks. Email filtering and anti-phishing training for employees can help to avoid phishing attacks, as well as the use of multi-factor authentication (MFA) to protect against credential theft.

Man-in-the-Middle (MitM) Attacks- An attacker intercepts communication between two parties to steal or manipulate data. Data theft, session hijacking, or credential compromise are a result of MitM attacks. They are avoided through the encryption of sensitive communications with SSL/TLS (HTTPS) or the use of VPNs (Virtual Private Networks) for secure remote access.

Malware & Ransomware Attacks- Malicious software infects a network to steal, delete, or encrypt data for ransom. Malware and ransomware attacks result in the loss of critical files, operational downtime, and financial extortion. Endpoint protection and antivirus software can be deployed to mitigate these attacks. Regular data backup and network segmentation is also helpful to prevent malware spread.

2.3 Security Mechanisms & Defensive Strategies

Firewalls- Block unauthorized traffic based on security rules.

Types:

- Network Firewalls – Protect networks at the perimeter (Cisco ASA, Palo Alto).
- Host-based Firewalls – Protect individual devices (Windows Defender Firewall).

Encryption- Converts data into unreadable formats to protect confidentiality.

Examples:

- SSL/TLS (Secure Sockets Layer) – Encrypt web traffic (HTTPS).
- AES (Advanced Encryption Standard) – Encrypt stored and transmitted data.

Intrusion Detection & Prevention Systems (IDS/IPS)- Detects and blocks suspicious activity within a network.

Examples:

- IDS (Intrusion Detection System) – Monitors traffic for threats (Snort, Suricata).
- IPS (Intrusion Prevention System) – Proactively blocks malicious traffic.

VLAN Segmentation- Segregates network traffic to limit unauthorized access. VLAN segmentation reduces attack surface and enhances network performance.

(GeeksforGeeks, n.d.)

2.4 Content Filtering & Compliance with IL and Federal Laws

Content Filtering- Blocks access to harmful, malicious, or non-compliant websites.

Examples:

- DNS Filtering – Blocks malicious domains (Cisco Umbrella, OpenDNS).
- Web Proxy Filtering – Prevents access to restricted websites (Squid Proxy).

Illinois & Federal Regulations

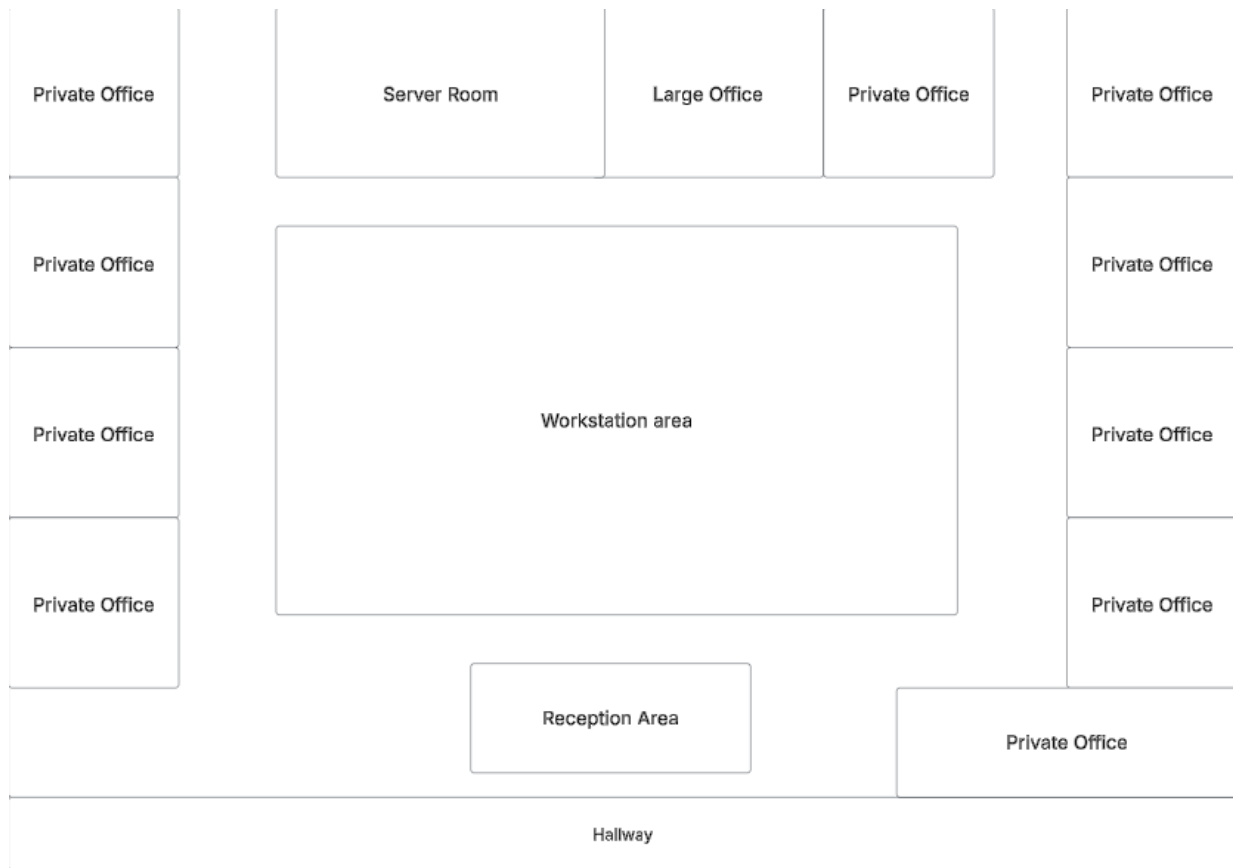
- Illinois Personal Information Protection Act (PIPA) – Requires encryption of sensitive data.
- Biometric Information Privacy Act (BIPA) – Protects biometric data from unauthorized use.
- GLBA (Gramm-Leach-Bliley Act) – Regulates financial data security.
- CISA (Cybersecurity Information Sharing Act) – Encourages organizations to report cyber threats.

3. Network Design and Proposal

3.1 Physical Layout Overview

Each tutoring center has a unique layout, requiring customized network designs:

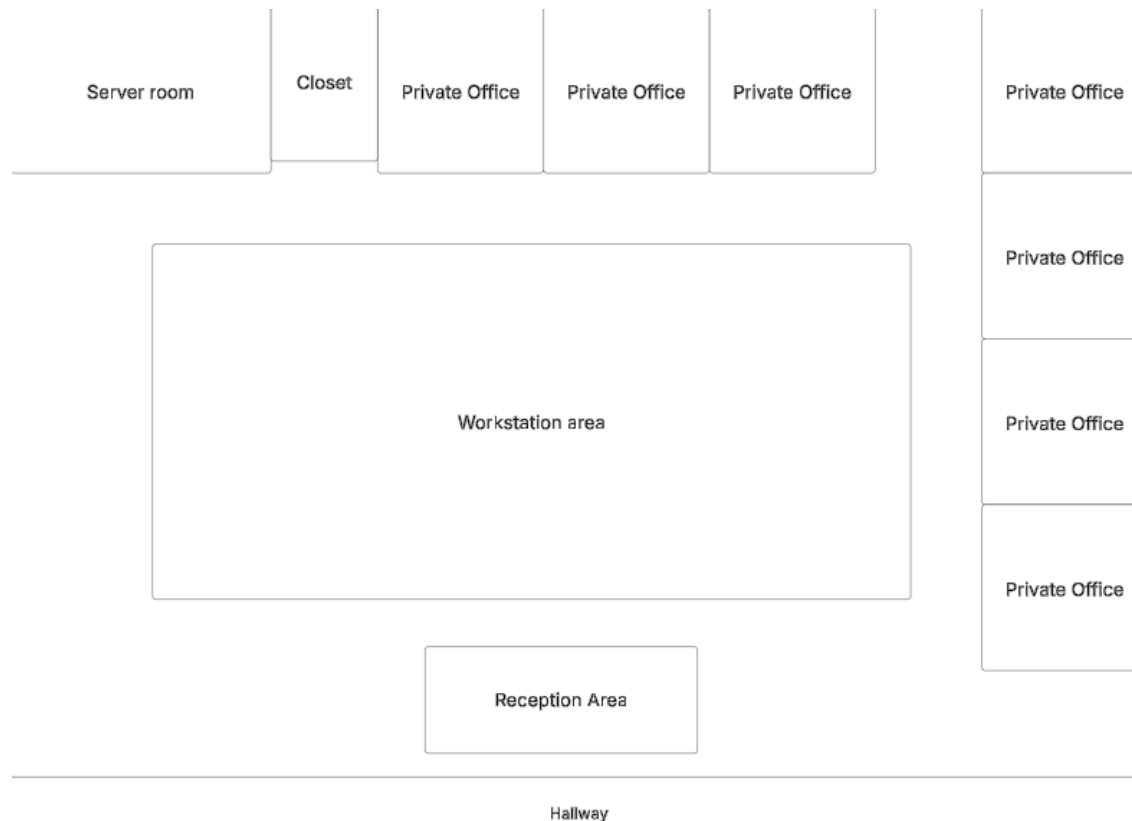
Romeoville, IL (Main Center - 3,600 sq ft): Reception area (400 sq ft), 1 large office (200 sq ft), 10 private offices (150 sq ft each), open workspace area (1200 sq ft), and a server room (400 sq ft)



Chicago, IL (3,000 sq ft): Reception area (400 sq ft), 11 private offices (150 sq ft each), open workspace area (600 sq ft), server room (150 sq ft), and two closets (100 sq ft each)



Aurora, IL (3,000 sq ft): Reception area (400 sq ft), 7 private offices (150 sq ft each), open workspace area (1200 sq ft), server room (200 sq ft), and a closet (150 sq ft)



(The final report will include a more detailed network diagram and floor plan made with LucidChart or draw.io, based on the instructor's feedback.)

3.2 Physical Topology and Installation Considerations

Wiring Runs

Romeoville (Main Center)

- **Drop Ceilings & Wall Pass-Throughs:** All data (Cat6a) and power cables will be run above the drop ceilings and through wall pass-throughs wherever necessary, ensuring minimal disruption to the tutoring center's operations.
- **Server Room Wiring Trunk:** A fiber backbone (plenum-rated if in ceiling spaces) connects the router/firewall to the core switch. Additional Cat6a cables fan out from the core switch to the reception area, private offices, and open workspace to support workstations, IP phones, and security cameras.

- Scheduled Off-Peak Installations: All new cable pulls and terminations occur after hours or during maintenance windows, reducing interruptions for staff and students.

Chicago

- Combination of Ceiling & Closet Runs: Uses plenum-rated fiber to connect the router/firewall to the main switch in the server room, running through ceiling spaces. Cat6a cables extend from the main switch to individual offices, the open workspace, and closets for wireless access point (WAP) installation.
- Local Code Compliance: All cable pathways, supports, and labeling meet local building codes and TIA/EIA standards for structured cabling.
- Minimized Downtime: Where feasible, existing conduits and cable trays are re-used to limit construction and reduce downtime.

Aurora

- Raised Floor Infrastructure: Uses riser-rated fiber under the raised floor to link the router/firewall with the core switch. Cat6a cables branch out from the core switch to each private office, reception area, and open workspace.
- Secure Cable Management: Cables are run in conduit or cable trays beneath the raised floor, protecting them from accidental damage and preserving a neat installation.
- Maintenance Windows: Installations and any re-routing are scheduled during off-peak hours to minimize impact on tutoring operations.

Additional Wiring Notes

- Safety & Building Codes:
 - All wiring follows national (NEC) and local regulations for fire safety, grounding, and cable ratings (plenum or riser, as applicable).
- Testing & Certification:
 - Each cable run (fiber or copper) is tested and certified to ensure proper signal integrity and performance before going live.
- Documentation & Labeling:
 - All cable endpoints and patch panels are labeled according to room/office designations and connected switch ports.

Equipment Relocation/Disposal: Legacy devices or wiring will be safely relocated, and outdated equipment will be recycled/disposed of following environmental, health, safety, and privacy guidelines. Disposal and relocation of electronic equipment and wiring are regulated by a variety of federal, state, and local laws designed to protect the environment, public health, and sensitive data. Laws to consider:

- Resource Conservation and Recovery Act (RCRA): This federal law governs the disposal of hazardous waste. Many legacy devices contain hazardous substances (such as lead, mercury, or cadmium) that must be disposed of in accordance with RCRA requirements.
- Occupational Safety and Health Administration (OSHA) Guidelines: OSHA provides standards to ensure safe work practices during the relocation or disposal of equipment, minimizing risks to workers (for example, safe removal of wiring and handling of heavy equipment).
- Privacy Regulations: Regulations such as the Federal Trade Commission (FTC) guidelines require secure data destruction for devices that store sensitive or personal data. This means that before disposal or recycling, all data should be properly erased or destroyed to protect privacy.
- Illinois' regulations under the Illinois Environmental Protection Act and the Illinois Solid Waste Management Act (40 ILCS 180) govern the proper handling and disposal of hazardous waste. Since many electronic devices contain hazardous substances (e.g., lead, mercury, cadmium), they must be disposed of through approved channels.

Physical Security: Critical devices (routers, switches, servers, firewalls) will be installed in locked server rooms or secure racks with proper environmental controls.

- Critical network devices—including routers, switches, servers, and firewalls—are housed in dedicated server rooms. These rooms are secured with electronic locking systems such as keycard access, PIN codes, or biometric scanners. Only authorized personnel are granted entry, ensuring that sensitive equipment is protected from unauthorized access.
- Within the secure server rooms, all critical devices are installed in lockable racks or cabinets. These enclosures come with built-in mechanical locks on their doors, so even if someone gains access to the room, they cannot tamper with the equipment without the proper keys or codes.
- Wireless Access Points (WAPs) will be mounted in strategic locations to ensure full coverage.

(Datar, 2023)

Each tutoring center (Romeoville, Chicago, and Aurora) leverages fiber cabling for its backbone connectivity. Fiber optic cables are used for:

- Core uplinks: Between the Internet edge (router/firewall) and the centrally managed switch.
- Inter-device connections: Connecting key servers, switches, and uplink ports in areas such as drop ceilings (using plenum-rated fiber) or under raised floors (using riser-rated fiber).
- Copper drops to workstations, security cameras, and wireless access points.
 - Cat6 Cables

(The Fiber Optic Association n.d.)

Romeoville

- 3,000 ft of Cat6a (for workstations, Wi-Fi APs, security cameras).
- 100-300 ft of fiber
- Extra patch cables 100ft of Cat6a

Chicago and Aurora

- 2,500 ft of Cat6a Ethernet
- 100-200 ft of fiber
- Extra patch cables 100 ft. of Cat6a

LAN Architecture

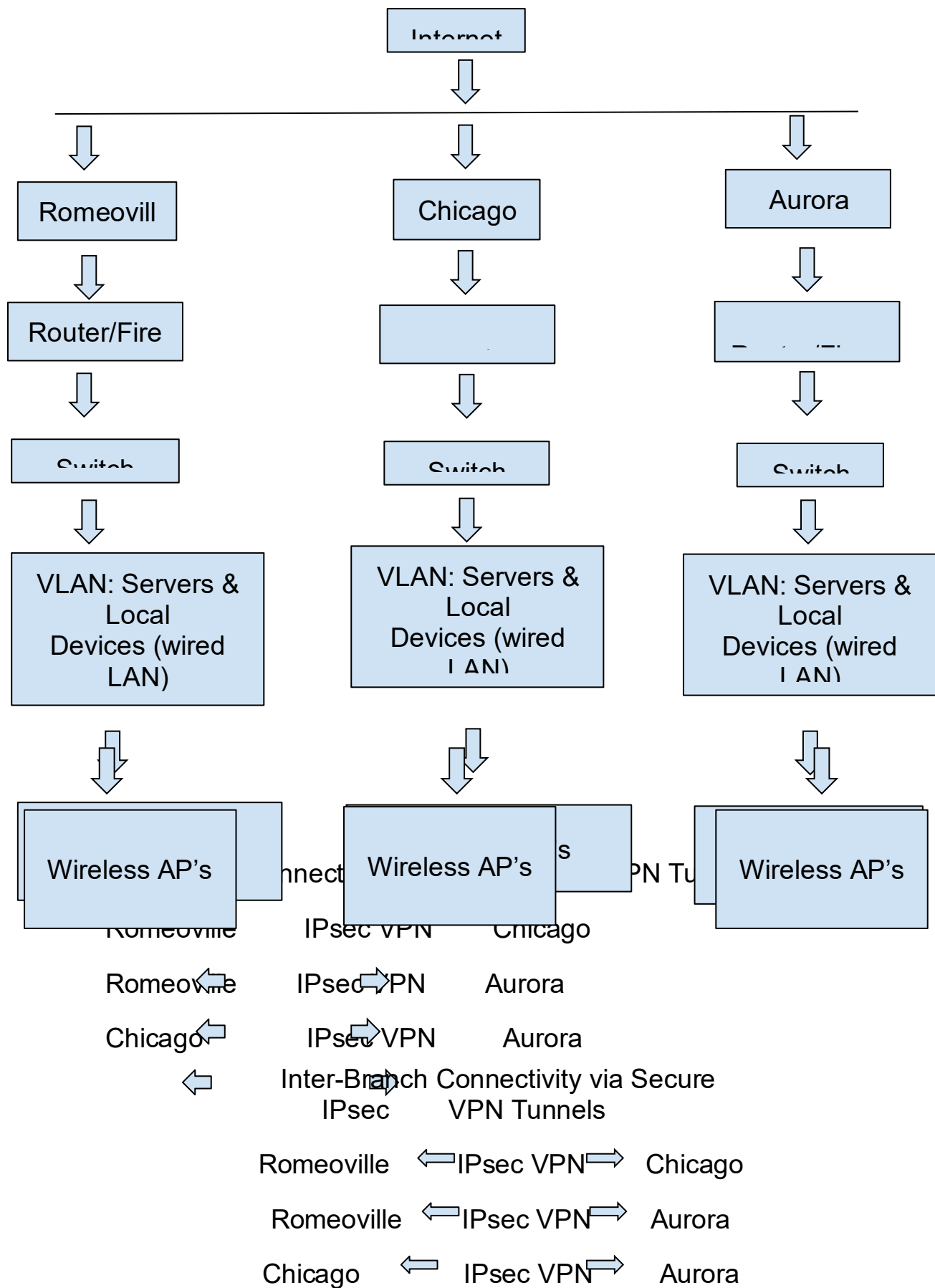
Internet Entry & Perimeter Devices

- Internet Connection:
 - The network begins with the Internet, which provides a dedicated high-speed connection to each tutoring center.
- Router/Firewall:
 - Installed in a secure server room, this device has fiber-enabled uplink ports that connect to the core network via fiber. Terminates the Internet connection, secures inbound/outbound traffic, and establishes IPsec VPN tunnels with the other branches (CDW, n.d.).
- Managed Switch:
 - Receives traffic from the router/firewall and distributes it within the branch, supporting VLAN segmentation for better network organization (CDW n.d.).
 - The switch is equipped with fiber ports. Fiber optic uplinks run from the router/firewall to the managed switch.
 - For Romeoville & Chicago (Drop Ceiling):
 - Use plenum-rated fiber optic cables (multimode or singlemode) run in the ceiling spaces (The Fiber Optic Association, n.d.).
 - For Aurora (Drop Floor):
 - Use riser-rated fiber optic cables routed under the raised floor for core uplinks (The Fiber Optic Association, n.d.).

Distribution and Local Access

- Server & Data Connectivity:
 - The switch segments the network into VLANs; one common VLAN carries the traffic for servers, workstations, and other local devices.
 - Core servers and storage devices are connected to the switch via fiber cables (if supported by the server NICs or via fiber media converters), ensuring high-speed data transfers.
- Wireless Access Points (WAPs):
 - Provide wireless connectivity across the branch, enabling mobile device access.
 - WAPs connect via copper Ethernet.
- End-User Devices:

- Workstations and IP phones are often connected via copper.
- Inter-Branch Connectivity:
 - Secure IPsec VPN tunnels interconnect the routers/firewalls at Romeoville, Chicago, and Aurora, ensuring that shared resources (databases, documents, and applications) can be accessed securely across locations.



3.3 LAN Flow

Romeoville Tutoring Center

1. Internet Connection:
 - A dedicated fiber/cable link enters the building.
2. Router/Firewall:
 - The FortiGate device terminates the Internet connection with fiber uplink modules.
3. Core Fiber Uplink:
 - A fiber cable runs from the router/firewall (using plenum-rated fiber in the drop ceiling) to the centrally located FortiSwitch.
4. VLAN Segmentation & Server Connectivity:
 - The managed switch uses fiber uplinks to connect to the switch, then local servers and storage devices.
5. Wireless & End-User Distribution:
 - Either copper drops or additional fiber connections (with media converters) are used to connect WAPs and wired devices as required.

Chicago Tutoring Center

1. Internet Connection:

- A dedicated link terminates at the site, connecting via fiber to the FortiGate router/firewall.
- 2. Fiber Backbone:
 - Plenum-rated fiber cables are used to connect the router/firewall to the FortiSwitch.
- 3. Internal Distribution:
 - The switch distributes traffic to the other switches with fiber links to key servers and optional copper for desktop drops.
- 4. Wireless Access:
 - WAPs connect to the switch; the backhaul may use fiber if desired, or standard copper ports.

Aurora Tutoring Center

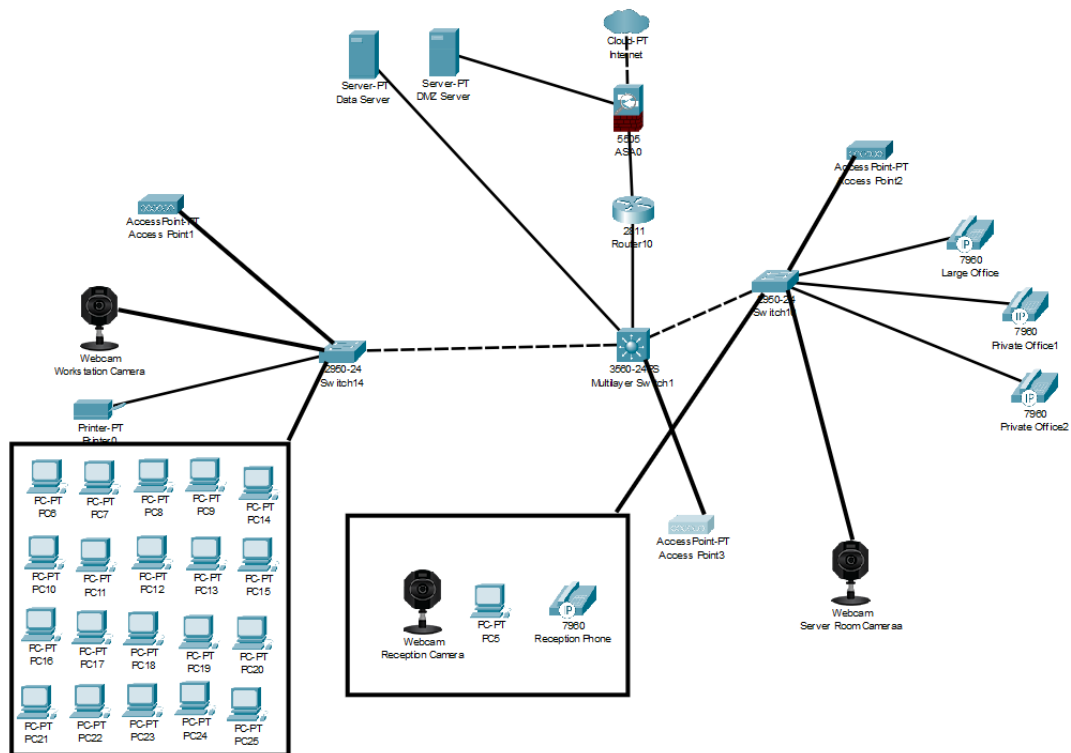
- 1. Internet Connection:
 - A high-speed link is brought in and terminated at a fiber-capable FortiGate router/firewall.
- 2. Fiber Under the Raised Floor:
 - Riser-rated fiber optic cables connect the router/firewall to the FortiSwitch, running beneath the raised floor.
- 3. Local Distribution:
 - The managed switch uses the fiber backbone to interconnect other switches, then local servers, storage, and distribution points.
- 4. Wireless & User Access:
 - Wireless and wired connections are provided; the critical uplink traffic is carried over fiber for maximum performance.

3.4 Physical Installation Guidelines

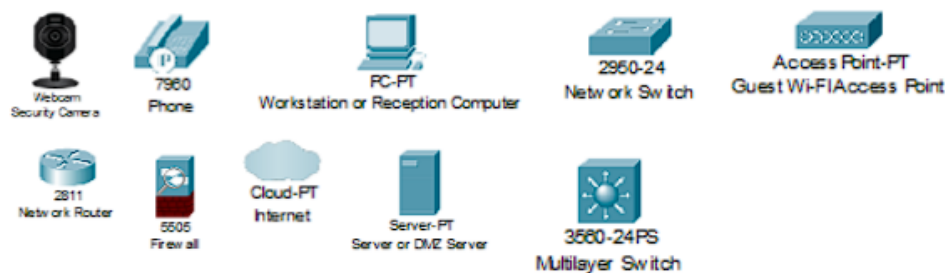
- Pre-Deployment Surveys:
 - Detailed site surveys in Romeoville, Chicago, and Aurora ensure that wiring (through walls, above ceilings, or under-floorings) is planned to minimize interference with business operations.
- Installation Scheduling:
 - Installations will be performed during off-peak hours or scheduled maintenance windows to avoid disruptions.
- Relocation and Disposal:
 - Any existing equipment is carefully relocated or recycled in compliance with environmental, health, safety, and privacy guidelines. STS Electronic Recycling offers free recycling and data destruction for computer, laptop, and electronics recycling, including networking equipment for dropoff. If pickup is needed, there would likely be a cost involved.
- Physical Security:

- All critical network devices are installed in locked server rooms or secure cabinets with proper cable management and environmental controls.

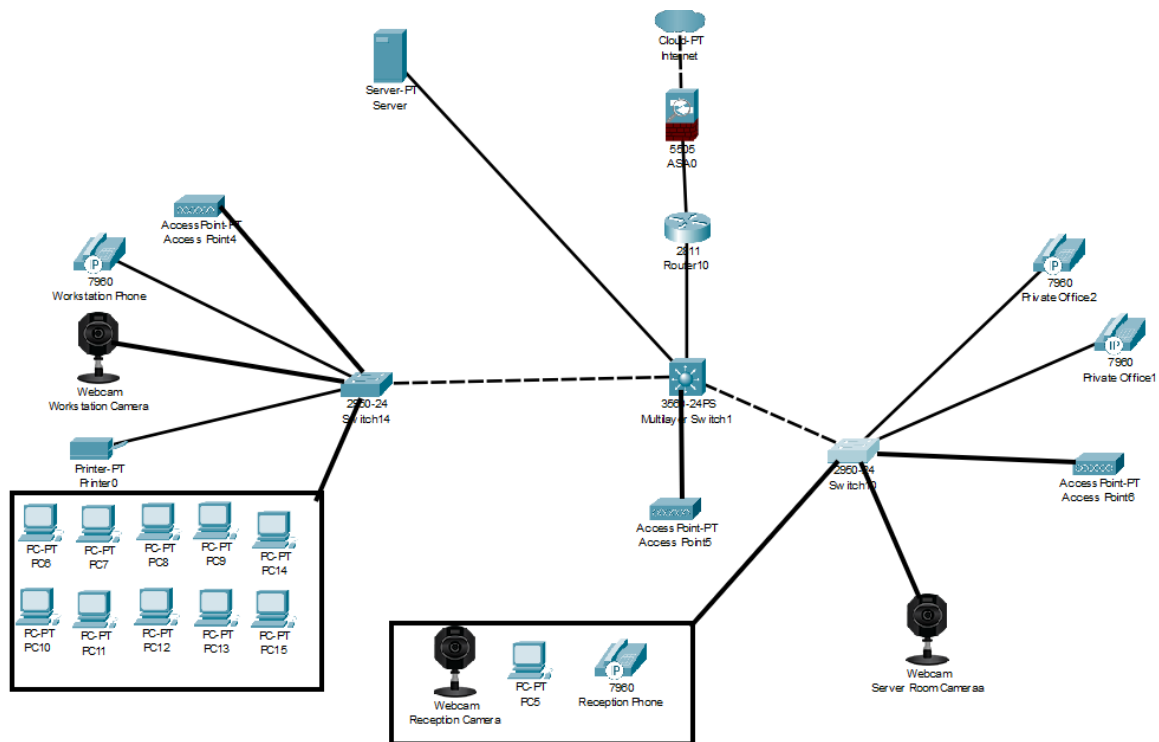
3.5 Packet Tracer Network Diagram



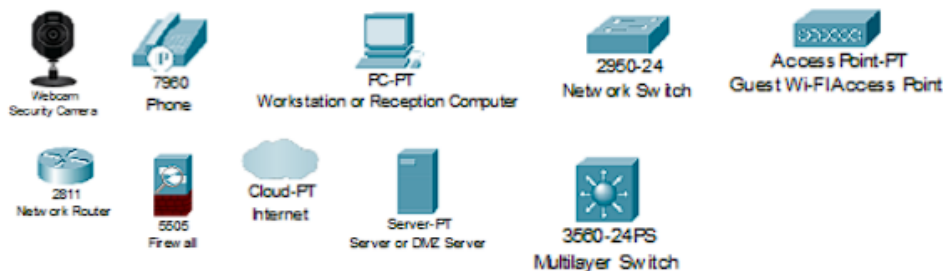
Packet Tracer Legend



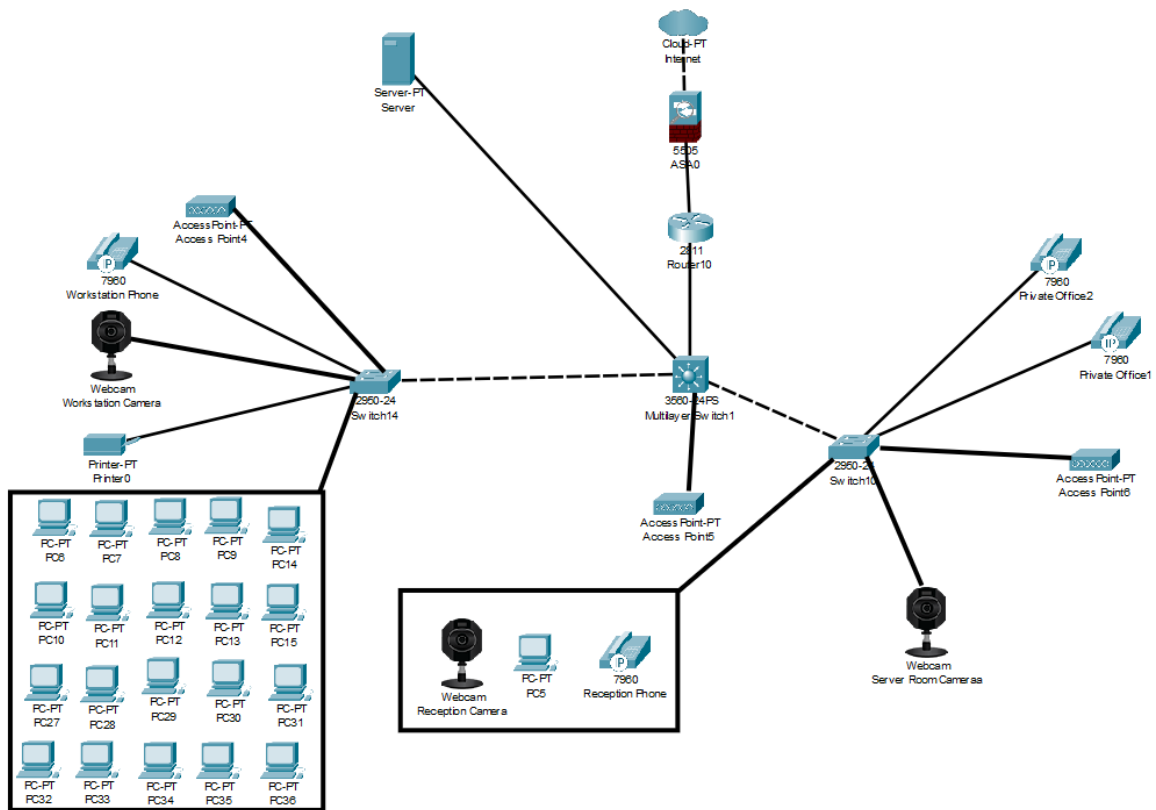
Romeoville Network: This location will have two switches, one will connect the devices in the open workstation area and the other will connect the devices from the reception area and the private offices. A multilayer switch will act as an intermediary for the two switches and will be connected to the router. This tutoring center will also have three wireless access points around the building for guests to receive a wireless connection without compromising the internal network. This location will also have a DMZ server connected to the firewall, to control and filter between the outside and internal networks.



Packet Tracer Legend



Chicago Network: This location will have two switches, one will connect the devices in the open workstation area and the other will connect the devices from the reception area and the private offices. A multilayer switch will act as an intermediary for the two switches and will be connected to the router. This tutoring center will also have three wireless access points around the building for guests to receive a wireless connection without compromising the internal network.



Packet Tracer Legend



Aurora Network: This location will have two switches, one will connect the devices in the open workstation area and the other will connect the devices from the reception area and the private offices. A multilayer switch will act as an intermediary for the two switches and will be connected to the router. This tutoring center will also have three wireless access points around the building for guests to receive a wireless connection without compromising the internal network.

3.6 Network Equipment Installation

Each location requires core networking equipment to support secure and reliable operations. Below is the recommended setup:

Location	Key Equipment
Romeoville	DMZ Server, Storage Server, Firewall, Router, Switches, Wireless APs, UPS
Chicago	Firewall, Router, Switch, Wireless APs, UPS
Aurora	Firewall, Router, Switch, Wireless APs, UPS

Network Equipment

DMZ Server:

1 (installed in Romeoville). A Demilitarized Zone (DMZ) Server is a security buffer that sits between an organization's internal network and external networks, such as the Internet. It is designed to host public-facing services while protecting internal resources from potential external threats (Geeks for Geeks, n.d.).

- Recommendation: Dell PowerEdge R750 Rack Server \$4,675.00. Supports high-performance processing and remote management. Customizable with different operating systems (CDW, n.d.).

Storage Server:

1 (installed in Romeoville). A storage server is a dedicated server designed to store, manage, and provide access to large amounts of data over a network. It acts as a central repository for files, databases, backups, and applications, ensuring data availability and security (Geeks for Geeks, n.d.).

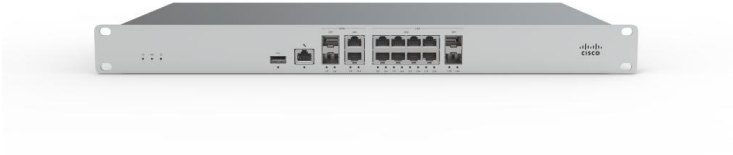
- Recommendation: Synology RackStation RS3621RPxs \$5000. Compatible with Dell DMZ, Cisco, and Juniper devices. 12-bay rackmount NAS (Network-Attached Storage). Comprehensive backup solutions (CDW, n.d.).

Networking Devices:

Routers:

3 (1 per location)- provides internet access, manages network traffic, and improves security (Geeks for Geeks, n.d.).

1. Cisco Meraki MX85



(CDW, n.d.)

The Cisco Meraki MX85 is a cloud-managed security and SD-WAN appliance designed for mid-sized branches. It offers integrated Layer 3 routing, firewall protection, and high-performance site-to-site VPN capabilities. Through the Meraki Dashboard, administrators can easily configure, monitor, and maintain the device remotely. Meraki's solution emphasizes ease of deployment and centralized management, ideal for distributed tutoring centers with minimal on-site IT resources.

Key Features:

- Firewall Throughput: 1 Gbps
- VPN Throughput: 500 Mbps
- Concurrent VPN Tunnels: 250+
- Cloud-Managed Dashboard
- Built-in Auto VPN for easy multi-site setup
- Integrated Layer 3 Routing and SD-WAN

Pros:

- Extremely easy to deploy and manage remotely (no CLI required)
- Auto-updating firmware and security signatures
- Centralized dashboard management for all locations
- Instant site-to-site VPN setup

Cons:

- Requires annual subscription/license for full features (security, SD-WAN)
- Limited customization for complex routing scenarios
- Dependent on Meraki Cloud availability

[Cisco Meraki Video](#)

2. Juniper SRX345



(CDW, n.d.)

The Juniper SRX345 is a high-performance security appliance that combines advanced firewall protection with sophisticated routing capabilities. Supporting full Layer 3 dynamic routing (OSPF, BGP, RIP) and flexible VPN configurations, the SRX345 is ideal for environments needing robust networking combined with detailed security policies. Though it offers greater customization and power than simpler routers, it typically requires a more experienced network administrator to configure properly.

Key Features:

- Firewall Throughput: 5 Gbps
- VPN Throughput: 800 Mbps
- Concurrent Sessions: 375,000+
- Layer 3 Dynamic Routing Protocols: OSPF, BGP, RIP
- Flexible VPN options (IPSec, GRE tunnels)
- Advanced NAT, policy-based routing

Pros:

- Superior advanced routing options compared to Meraki
- Strong integrated firewall and VPN performance

- Extensive CLI and scripting support for customization
- No forced cloud subscription required

Cons:

- More complex setup and administration (steeper learning curve)
- Requires skilled network engineers for optimal configuration
- Limited built-in GUI compared to Meraki

Summary: Juniper SRX345 provides deep routing capabilities, high VPN flexibility, and have skilled network administrators available to manage it.

[Juniper SRX345 Video](#)

3. Fortinet FortiGate 100F



(CDW, n.d.)

The Fortinet FortiGate 100F functions as both a router and a next-generation firewall, providing fast Layer 3 routing capabilities, high-speed VPN tunneling, and Unified Threat Management (UTM). It supports OSPF, BGP, and RIP protocols for dynamic routing, making it ideal for multi-site tutoring centers needing strong connectivity and security. Fortinet's FortiOS operating system enables detailed traffic control, VPN management, and automated threat protection from a single interface.

Key Features:

- Firewall Throughput: 20 Gbps
- VPN Throughput: 11 Gbps
- Concurrent Sessions: 2 Million
- Layer 3 Routing Protocols: OSPF, BGP, RIP supported

- Unified Threat Management (UTM): antivirus, web filtering, IPS built-in
- Built-in SD-WAN and advanced VPN

Pros:

- Best cost-to-performance ratio among the three
- Handles both firewall + router roles efficiently
- High-speed VPN connectivity across sites
- Comprehensive security and routing features in one platform

Cons:

- GUI slightly less intuitive than Meraki Dashboard
- Advanced features (sandboxing, extended threat protection) require extra licensing
- Learning curve for full FortiOS capabilities

[Fortinet Fortigate 100F Video](#)

Recommendation:

Fortinet FortiGate 100F provides the strongest combination of security, routing performance, and value. Will give the tutoring centers high-speed VPN tunnels, deep protection, and future scalability (CDW, n.d.).

Switches:

Each location will have multiple user groups that will need VLANs for security and traffic management. A VLAN (Virtual Local Area Network) is a way to segment a physical network into multiple isolated networks without requiring separate physical hardware- proposing 9 total switches (3 at each location). (Geeks for Geeks, n.d.).

1. Cisco Meraki MS250-48P



(CDW, n.d.)

This is the Cisco Meraki MS250-48P cloud-managed switch. It features 48 Gigabit Ethernet ports with PoE+ support and 4 high-speed 10G SFP+ uplink ports. Administrators manage the MS250-48P through the Meraki Dashboard, enabling simple remote configuration, automatic firmware updates, and seamless integration into cloud-managed networks across multiple sites. Its design prioritizes ease of deployment and centralized monitoring, ideal for distributed environments like tutoring centers.

Key Features:

- Ports: 48 x 1G Ethernet ports + 4 x 10G SFP+ uplink ports
- PoE/PoE+ Support: Yes (740W available)
- Layer 3 Routing: Static Routing, DHCP Server, OSPF in beta
- Cloud Managed via Meraki Dashboard
- Stacking Support: Up to 8 switches with dedicated stacking ports
- Lifetime Warranty with advanced replacement

Pros:

- Simple cloud-based management (Meraki Dashboard)
- Automatic firmware and security updates
- High PoE budget — perfect for powering many APs and IP phones
- Great stacking options for larger networks

Cons:

- Expensive (both hardware and licensing fees)
- Dependent on cloud licensing for full features

- Limited Layer 3 functionality compared to standalone enterprise switches

Summary: Choose Cisco Meraki MS250-48P if you prioritize cloud simplicity, easy scaling, and remote management, and you have budget for ongoing licensing.

[Cisco Meraki Video](#)

2. Fortinet FortiSwitch 148E-POE



(CDW, n.d.)

This image shows the Fortinet FortiSwitch 148E-POE, a secure, performance-focused Ethernet switch offering 48 PoE-capable ports. It integrates tightly with FortiGate firewalls via FortiLink, allowing unified security and network management from a single console. The FortiSwitch 148E-POE supports VLAN segmentation, QoS traffic prioritization, and security fabric integration without requiring additional licensing fees, making it an excellent choice for high-security network deployments.

Key Features:

- Ports: 48 x 1G Ethernet ports + 4 x 1G SFP uplink ports
- PoE/PoE+ Support: Yes (370W budget)
- Layer 2+ Switching: VLANs, STP/RSTP, link aggregation
- Managed via FortiGate or FortiLink
- Seamless Integration with Fortinet Security Fabric
- No recurring cloud license needed

Pros:

- Affordable compared to Meraki
- Tight integration with FortiGate firewalls (one-click manageability)

- Strong PoE support for APs, IP phones, cameras
- No extra subscription needed for full management

Cons:

- No full Layer 3 routing (limited to VLAN routing)
- Not as fancy a GUI as Meraki Dashboard
- Configuration slightly more technical if standalone

Summary: Choose FortiSwitch 148E-POE if you want cost-effective, secure, high-performance switching that integrates perfectly with Fortinet routers/firewalls.

[Fortinet FortiSwitch 148E-POE Video](#)

3. Juniper EX3400-48P



(CDW, n.d.)

The Juniper EX3400-48P, an enterprise-grade switch with 48 PoE+ Gigabit ports and 4 10G uplink ports. It offers full Layer 2 and Layer 3 dynamic routing support, including OSPF, BGP, and VRRP protocols. Featuring Virtual Chassis technology, up to 10 EX3400 switches can operate as a single logical device, simplifying management. Its advanced QoS, MACsec encryption, and security features make it highly suitable for larger, resilient network infrastructures.

Key Features:

- Ports: 48 x 1G Ethernet ports + 4 x 10G SFP+ uplink ports
- PoE/PoE+ Support: Yes (740W available)
- Full Layer 2 and Layer 3 Routing: OSPF, BGP, VRRP, PIM
- Virtual Chassis Technology — stack up to 10 switches

- Junos OS Enterprise Operating System
- Advanced QoS, MACsec encryption support

Pros:

- Real Layer 3 dynamic routing capabilities
- High resilience and enterprise-class performance
- Powerful QoS for voice/video prioritization
- Advanced security and compliance features

Cons:

- Most expensive of the three
- Requires Junos OS knowledge (higher admin skill needed)
- Configuration complexity not ideal for smaller IT teams

Summary: Juniper EX3400-48P only if your tutoring centers need advanced Layer 3 routing, super high network reliability, and you have experienced network admins to manage Junos devices.

[Juniper EX3400-48P Video](#)

Recommendation:

Fortinet FortiSwitch 148E-POE is recommended as the primary switching solution. It provides a strong balance of security, PoE support, affordability, and seamless integration with FortiGate firewalls without requiring ongoing licensing costs.

Wireless Access Points:

9 units (3 per location). A Wireless Access Point (WAP) is a networking device that extends a wired network by providing wireless connectivity. It allows Wi-Fi-enabled devices to connect to a local area network (LAN) and access the internet (Geeks for Geeks, n.d.).

1. Cisco Meraki MR46



(CDW, n.d.)

This is the Cisco Meraki MR46, a cloud-managed Wi-Fi 6 access point with 4x4 MU-MIMO antenna arrays. It offers high-throughput wireless performance, VLAN-ready SSID configurations, and centralized management through the Meraki Dashboard. Designed for simple multi-site deployment, it ensures automatic firmware updates, client traffic optimization, and seamless integration into cloud-based networks across the tutoring centers.

Key Features:

- Wi-Fi Standard: Wi-Fi 6 (802.11ax)
- Max Data Rate: Up to 3.5 Gbps
- Antennas: 4x4 MU-MIMO
- Management: 100% Cloud-managed via Meraki Dashboard
- VLAN Support: Yes (SSIDs map to VLANs)
- Integrated Security: Auto VPN, RF optimization, built-in firewall

Pros:

- Fully cloud-managed with easy setup
- Strong Wi-Fi 6 performance for high-density environments
- Automatic firmware and security updates
- Seamless VLAN tagging per SSID

Cons:

- Requires annual licensing subscription
- Higher cost compared to Fortinet or Aruba alternatives

- Dependent on cloud service availability

Summary: Cisco Meraki MR46 prioritizes cloud-managed simplicity, seamless VLAN integration, and enterprise-grade Wi-Fi 6 performance, and are comfortable with recurring subscription costs.

[Cisco Meraki Video](#)

2. Aruba AP-515



(CDW, n.d.)

This image shows the Aruba AP-515, an enterprise-grade Wi-Fi 6 access point that supports AI-based radio optimization for intelligent channel and power management. The AP-515 is capable of achieving multi-gigabit wireless speeds and offers flexible deployment options, including controller-based or cloud-managed configurations via Aruba Central. It is well-suited for high-density environments like tutoring centers that demand reliable and optimized wireless coverage.

Key Features:

- Wi-Fi Standard: Wi-Fi 6 (802.11ax)
- Max Data Rate: Up to 5.37 Gbps
- Antennas: 4x4 MU-MIMO
- AI Optimization: Intelligent RF tuning and client steering
- VLAN Support: Yes
- On-premises or Cloud Management Options (Aruba Central optional)

Pros:

- High-speed enterprise-grade Wi-Fi 6 performance
- AI-based radio optimization for best coverage

- Flexible deployment (cloud-managed or controller-managed)
- Lower cost than Cisco Meraki

Cons:

- Advanced features require Aruba Central licensing
- Slightly more complex initial configuration than Meraki
- Might require additional tuning for smaller networks

Summary: Aruba AP-515 provides very high wireless performance, AI-optimized RF management, and the option of cloud or on-premises control without fully committing to cloud-only management.

[Aruba AP-515 Video](#)

3. Fortinet FortiAP 231F



(CDW, n.d.)

Key Features:

- Wi-Fi Standard: Wi-Fi 6 (802.11ax)
- Max Data Rate: Up to 2.5 Gbps
- Antennas: 2x2 MU-MIMO (per band)
- Management: FortiGate Firewall Controller or FortiCloud
- Auto-configuration: Auto-integrates with FortiGate and FortiSwitch
- Built-in Security: Directly integrates into Fortinet Security Fabric

Pros:

- Lowest cost option with full Wi-Fi 6 support
- Automatic integration with FortiGate firewalls and FortiSwitches
- No additional licensing required for basic management
- Centralized security and wireless policy enforcement

Cons:

- Management experience is best when using FortiGate (tight integration)
- Slightly lower max throughput compared to Aruba AP-515
- More technical configuration without FortiGate controller

[Fortinet Fortigate 100F Video](#)

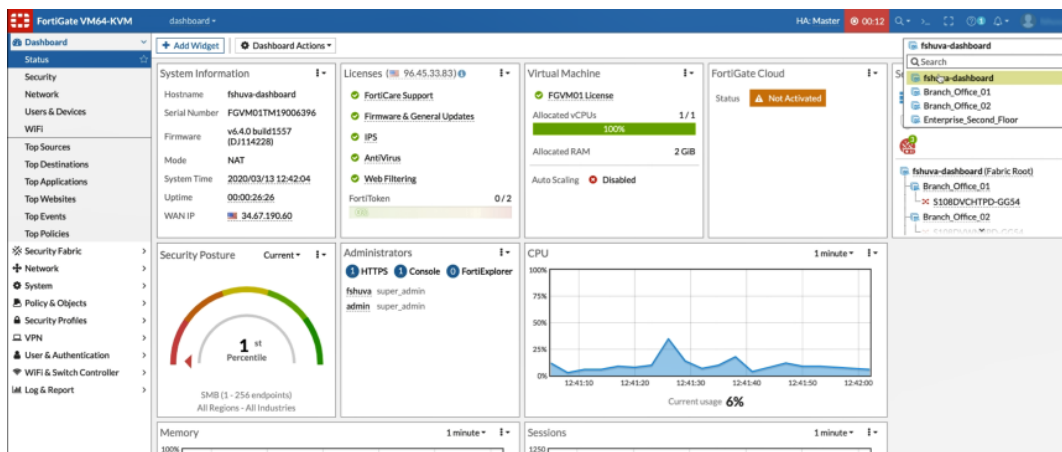
Recommendation:

Fortinet FortiAP 231F will offer the easiest auto-configuration, tightest security integration, and best cost efficiency without recurring cloud license costs.

Firewall:

3 (1 per location). A firewall is a security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules. (Geeks for Geeks, n.d.)

1. Fortinet FortiGate 100F



This is the administrative dashboard for the Fortinet FortiGate 100F. It provides an overview of firewall performance, bandwidth usage, active VPN tunnels, intrusion prevention system (IPS) alerts, and real-time system health. Administrators can quickly identify network security threats, monitor VPN activity between tutoring centers, and verify device resource loads (CPU, memory) from a single centralized page.

Key Features:

- Firewall Throughput: 20 Gbps
- Threat Protection Throughput: 1.2 Gbps
- Concurrent Sessions: 2 million
- IPsec VPN Throughput: 11 Gbps
- Integrated SD-WAN and NGFW (Next-Gen Firewall)

Pros:

- Cost-Effective for medium enterprises and branch offices
- Built-in UTM (Unified Threat Management): IPS, antivirus, application control
- High VPN performance — crucial for branch-to-branch tunneling
- Seamless integration with FortiSwitches and FortiAPs
- No mandatory cloud licensing unlike Meraki

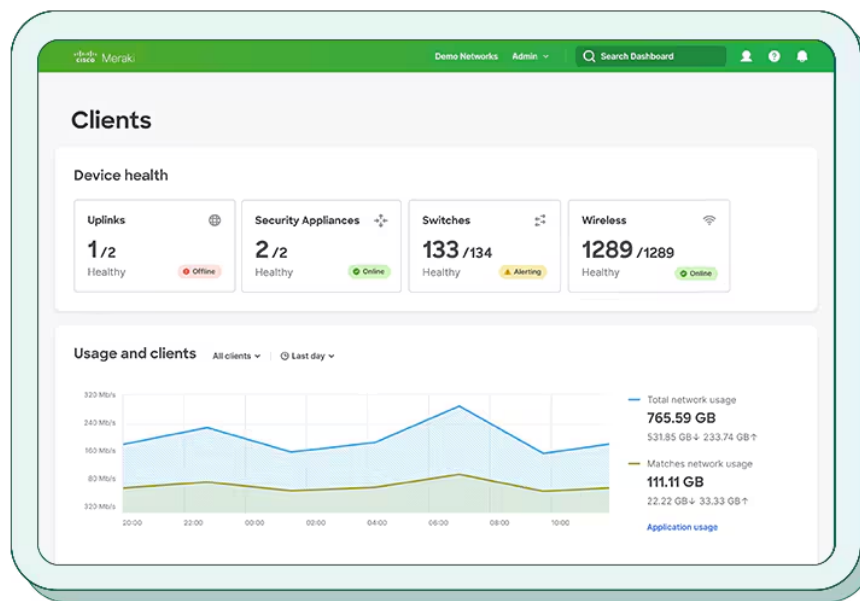
Cons:

- UI less user-friendly compared to Meraki (slight learning curve)
- Advanced features (e.g., sandboxing) require additional licenses
- Physical space: 1U rackmount size might be overkill for very small sites

Summary: Best balance of price, performance, and advanced security.

[Fortinet Fortigate 100F Video](#)

2. Cisco Meraki MX85



This is the Cisco Meraki MX85 cloud-based firewall management dashboard. It displays centralized control of firewall rules, VPN connections, traffic analytics, and device statuses across multiple sites. Through the Meraki Dashboard, IT administrators can remotely configure security policies, monitor real-time WAN performance, and troubleshoot network issues without needing on-site access to the physical device. Meraki's intuitive GUI is especially helpful for small teams managing multiple tutoring centers.

Key Features:

- Firewall Throughput: 1 Gbps
- Advanced Security License required for threat protection
- Concurrent Sessions: 500,000
- Integrated SD-WAN
- Cloud-based dashboard management

Pros:

- Simplified Cloud Management — Manage everything via Meraki Dashboard
- Auto VPN Setup — Instant setup for site-to-site VPNs
- Automatic firmware upgrades and threat signature updates
- Very easy to scale for multiple sites

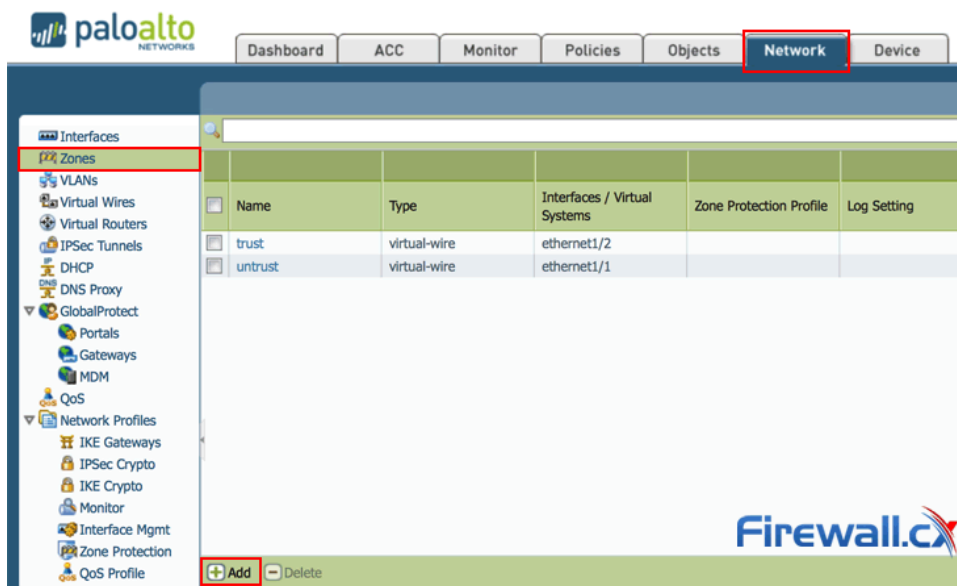
Cons:

- High recurring licensing cost (\$400/year+ per device for Advanced Security License)
- Performance bottlenecks under heavy loads (especially if IPS/IDS are enabled)
- Cloud dependency: Internet outage = limited local management

Summary: Best if you prioritize ease of management and don't mind annual license fees.

[Cisco Meraki Video](#)

3. Palo Alto Networks PA-220



Key Features:

- Firewall Throughput: 500 Mbps
- Threat Prevention Throughput: 300 Mbps
- Concurrent Sessions: 64,000
- App-ID and User-ID: Deep traffic identification features
- Zero Trust Networking capabilities built-in

Pros:

- Very powerful security stack (application-level, user-level, content inspection)

- Best-in-class threat intelligence (WildFire sandboxing)
- **Built-in support for Zero Trust security models
- High visibility into application usage and risks

Cons:

- More complex initial configuration (requires trained admin)
- Higher upfront and support costs than Fortinet
- Lower throughput meaning not suitable for sites needing high VPN bandwidth

Summary: Best if you prioritize advanced application-aware security but have strong IT administration available.

[Palo Alto Video](#)

Final Recommendation

For your Romeoville, Chicago, and Aurora tutoring centers:

- Primary Recommendation: Fortinet FortiGate 100F — best combination of cost, performance, VPN, and security.
- Alternate Recommendation (cloud managed, if preferred): Cisco Meraki MX85 (but factor license renewals!)
- Palo Alto PA-220 would only be recommended if you had very strict compliance or application filtering requirements — not needed for typical tutoring center operations.

Power Backup

- UPS (Uninterruptible Power Supply) is a backup power device that provides temporary electricity during power outages and protects equipment from power surges.: 3 units (1 per location for server room equipment). Likely to be included in the 20000 additional budget (TechTarget, n.d.).

Option APC Smart-UPS SMT1000C \$419.99 (CDW, n.d.)

Communication/Other Equipment

- Phones: 12 units (4 per location for reception, open workspace, and private offices). Cisco, Grandstream, Yealink \$50-150 depending on voice preference. As this is a smaller deployment with basic management needs, Yealink devices can often be configured and managed individually

without additional subscription services and may be the best option. If centralized management is needed, that can be deployed later with additional subscription costs (CDW, n.d.).

- Headsets: 6 units (2 per location for administrative or faculty use). Jabra Evolve 20 MS stereo \$48 (CDW, n.d.)
- Printers: 6 (2 per location)

Option: Canon imageCLASS MF743Cdw \$550 (CDW, n.d.)

- Cameras:
 - Wide-angle cameras: 12 units (3 per location for reception, hallways, server room, and open workspaces) \$56.99 (CDW, n.d.)
- Non-cash payments handled through Square POS System at each location. Square POS is a comprehensive point-of-sale solution that integrates hardware and software for in-person transactions. It features an intuitive tablet interface and dedicated card readers for processing payments, while also offering inventory management, sales tracking, and reporting capabilities—all designed to streamline retail or service operations without relying on a cloud-based infrastructure (Square, n.d.).

4. Logical Topology & Subnet Design

Each tutoring center (Romeoville, Chicago, Aurora) has:

- Separate subnets for: Staff, Tutors, Guests, Servers, and DMZ (Romeoville).
- Use VLANs to implement logical separation.
- IP Schema:

Location	VLAN Name	VLAN ID	Subnet	Gateway IP	Purpose
Romeoville	Staff	10	192.168.10.0/24	192.168.10.1	Admins & employees
Romeoville	Tutors	20	192.168.20.0/24	192.168.20.1	Tutor computers
Romeoville	Guests	30	192.168.30.0/24	192.168.30.1	Wi-Fi Internet only
Romeoville	Servers	40	192.168.40.0/24	192.168.40.1	AD, files, backup
Romeoville	DMZ	50	192.168.50.0/24	192.168.50.1	Web or external services
Chicago	Staff	10	192.168.60.0/24	192.168.60.1	Admins & employees
Chicago	Tutors	20	192.168.70.0/24	192.168.70.1	Tutor computers
Chicago	Guests	30	192.168.80.0/24	192.168.80.1	Wi-Fi Internet only
Chicago	Servers	40	192.168.90.0/24	192.168.90.1	File and print servers

Aurora	Staff	10	192.168.100.0/24	192.168.100.1	Admins & employees
Aurora	Tutors	20	192.168.110.0/24	192.168.110.1	Tutor computers

4.1 Machine Configuration (Routers, Servers, Workstations, Devices)

Each machine/device should be configured as follows:

Romeoville Tutoring Center

VLAN Subnets:

- Staff VLAN: 192.168.10.0/24
 - Gateway: 192.168.10.1
 - Devices (e.g., workstations, printers): 192.168.10.100–192.168.10.110
- Tutors VLAN: 192.168.20.0/24
 - Gateway: 192.168.20.1
 - Devices (e.g., tutors' workstations, printers): 192.168.20.100–192.168.20.110
- Guests VLAN: 192.168.30.0/24
 - Gateway: 192.168.30.1
 - Devices (e.g., guest access points): 192.168.30.100–192.168.30.110
- Servers VLAN: 192.168.40.0/24
 - Gateway: 192.168.40.1
 - Critical Servers (AD, DNS): 192.168.40.5
 - Additional Servers (file, print, backup): 192.168.40.10–192.168.40.20
- DMZ VLAN: 192.168.50.0/24
 - Gateway: 192.168.50.1
 - DMZ Server: 192.168.50.5
 - Other DMZ devices: 192.168.50.10–192.168.50.20

Physical Layout: Rooms include a reception area, a large office, 10 private offices, an open workspace, and a server room. Each VLAN's devices are placed as follows: critical infrastructure (servers, DMZ) are in the server room, while workstations, printers, and APs are distributed in offices and open workspaces.

Chicago Tutoring Center

VLAN Subnets:

- Staff VLAN: 192.168.60.0/24
 - Gateway: 192.168.60.1
 - Devices: 192.168.60.100–192.168.60.110
- Tutors VLAN: 192.168.70.0/24
 - Gateway: 192.168.70.1
 - Devices: 192.168.70.100–192.168.70.110
- Guests VLAN: 192.168.80.0/24
 - Gateway: 192.168.80.1
 - Devices: 192.168.80.100–192.168.80.110
- Servers VLAN: 192.168.90.0/24
 - Gateway: 192.168.90.1
 - Critical Servers: 192.168.90.5
 - Additional Servers: 192.168.90.10–192.168.90.20
- DMZ VLAN: 192.168.100.0/24
 - Gateway: 192.168.100.1
 - DMZ Server: 192.168.100.5
 - Other DMZ Devices: 192.168.100.10–192.168.100.20

Physical Layout: This location includes a reception area, 11 private offices, an open workspace, a server room, and two closets. Each VLAN's devices are assigned based on their function and location; for example, servers are in the server room while end-user devices (workstations, printers) are distributed throughout the office areas.

Aurora Tutoring Center

VLAN Subnets:

- Staff VLAN: 192.168.110.0/24
 - Gateway: 192.168.110.1
 - Devices: 192.168.110.100–192.168.110.110
- Tutors VLAN: 192.168.120.0/24
 - Gateway: 192.168.120.1
 - Devices: 192.168.120.100–192.168.120.110
- Guests VLAN: 192.168.130.0/24
 - Gateway: 192.168.130.1
 - Devices: 192.168.130.100–192.168.130.110
- Servers VLAN: 192.168.140.0/24
 - Gateway: 192.168.140.1
 - Critical Servers: 192.168.140.5

- Additional Servers: 192.168.140.10–192.168.140.20
- DMZ VLAN: 192.168.150.0/24
 - Gateway: 192.168.150.1
 - DMZ Server: 192.168.150.5
 - Other DMZ Devices: 192.168.150.10–192.168.150.20

Physical Layout: Aurora includes a reception area, 7 private offices, an open workspace, a server room, and a closet. As with the other centers, devices are grouped by VLAN based on function: servers and DMZ equipment in the server room, while workstations, printers, and APs are strategically placed across the facility guided by a WiFi heatmap and detailed floor plan.

Summary:

For each location, the network is divided into multiple VLAN-based subnets. Within each VLAN:

- The router (or gateway interface) is always assigned the .1 address.
- Critical servers are assigned .5 (or fall within .10–.20 for other servers).
- End-user devices such as switches, APs, and printers receive addresses in the .100–.110 range.

5. Equipment Selection and Cost Breakdown

5.1 [Recommended Network Infrastructure and Miscellaneous Costs](#)

Note: The Fortinet FortiGate 60F firewall requires a subscription for ongoing security services and support. Fortinet offers two bundles with a yearly fee that does vary in different locations/vendors: The Enterprise bundle yearly fee is around \$800 and offers comprehensive services such as IPS (Intrusion Prevention System), Advanced Malware Protection, Application Control, Web & Video Filtering, Antispam, Security Rating, IoT Detection, Industrial Security, FortiConverter Service, and 24x7 FortiCare support. The UTP (Unified Threat Protection) bundle which would be around \$500 is a paired-down version of the Enterprise but still offers essential services like IPS, Advanced Malware Protection, Application Control, Web & Video Filtering, Antispam, and 24x7 FortiCare support. The UTP would be sufficient for this current configuration (Fortinet).

6. ISP Recommendations

6.1 Internet Service Providers (ISP) Evaluation

- Verizon Business: Best overall reliability, but higher cost
- AT&T Business: Best for customer satisfaction
- Comcast Business: Best for availability

6.2 Recommended ISP Plan

- Romeoville & Chicago: Fiber-optic AT&T Business for high-speed, low-latency connectivity. Romeoville would likely benefit from the higher bandwidth rate.

- Internet 300: \$55 per month plus tax, offering 300 Mbps symmetrical upload/download speeds.
- Internet 500: \$65 per month plus tax, providing 500 Mbps symmetrical speeds.

Data and bandwidth requirements are typically calculated by estimating the number of users and devices, their usage patterns, and then adding a margin for peak traffic and future growth. Factors include:

- **Device Count & Usage:** Count of Devices: Identify the number of workstations, IP phones, wireless devices, and other connected equipment at each tutoring center. Concurrent Users: Estimate the number of devices that will be actively using the network simultaneously (e.g., during class or study sessions).
- **Application Requirements- Internet Activities:** Assess the primary uses (web browsing, video conferencing, file transfers, streaming educational content) to determine the typical bandwidth each activity consumes. Video Conferencing: Since many educational activities now include video, higher upload speeds are necessary to maintain quality.
- **Peak Traffic Estimation:- Usage Patterns:** Evaluate peak usage periods (e.g., during online classes or tutoring sessions) to determine maximum simultaneous demand.
- **Symmetrical Speeds-** Educational environments often need balanced upload and download speeds (symmetrical) to support both content consumption and creation (like video calls or uploading assignments).
- **Industry Guidelines-** Reference typical requirements for small-to-medium educational facilities and adapt estimates based on the specific number of users and devices.

The tutoring centers have 50 devices and each device requires on average 5 Mbps of bandwidth during peak periods (considering a mix of activities), that would suggest a baseline need of 250 Mbps. Adding overhead (say 20–30%) and planning for growth may then lead to choosing a plan offering 300 Mbps or 500 Mbps symmetrical speeds. This estimation process ensures that the tutoring centers have adequate capacity to handle both current needs and anticipated future demands.

- **Aurora:** Upgrading to AT&T Business Fiber can enhance Aurora's tutoring centers' connectivity. Upgrading to fiber cost would depend on whether complex construction would be needed for the upgrade. If the upgrade is standard, the installation cost would be low, likely less than \$300. If minor construction would be needed to lay fiber through an existing pathway, the cost could range from \$500-\$2000.

(AT&T, n.d.)

7. Security and Compliance Implementation

7.1 Security Devices for Network Protection

Firewalls:

- Function: Prevent unauthorized access and filter traffic at the network perimeter.
- Possible Solutions:
 - Fortinet FortiGate 60F- FortiGate integrates Unified Threat Protection (UTP), SD-WAN, and VPN services in one appliance.
 - Cisco ASA – Known for enterprise features and Cisco ecosystem integration.
 - Juniper SRX Series – Offers advanced routing, firewall, and VPN capabilities with strong performance.

Suggested Firewall rules:

- Allow VLAN-to-VLAN communication (as needed)
- Block Guest VLAN from internal access
- Allow outbound Internet from all VLANs
- DMZ only accessible from Internet and Servers VLAN

Intrusion Detection & Prevention Systems (IDS/IPS)

- Function: Detect and block suspicious activities that could compromise network security.
- IDS/IPS solutions monitor network traffic in real time, identifying known attack patterns and stopping threats before they spread.
- Possible Solutions:
 - Snort (open-source), Suricata (open-source), FortiGate IPS (integrated with Fortinet appliances).

Security Information and Event Management (SIEM)

- Function: Aggregate logs and security events from multiple devices, perform real-time analysis, and alert administrators of threats.
- SIEM solutions provide centralized visibility, correlation of events, and compliance reporting, crucial for a multi-location tutoring center.
- Potential Tools:
 - Splunk Enterprise Security, IBM QRadar, Microsoft Sentinel.

LAN Segmentation

- Function: Isolate networks to limit security risks and reduce the attack surface.

- Implementation: Separate VLANs for staff, tutors, students, and guests.
- Benefit: Even if one VLAN is compromised, lateral movement is restricted.

Site-to-Site VPNs

- Function: Encrypt communication between different tutoring center locations (Romeoville, Chicago, Aurora).
- Benefit: Ensures secure data transfer over the public internet, maintaining confidentiality and integrity.

Web Filtering & Content Compliance

- Function: Enforce compliance with Illinois and federal regulations (K-12 filtering requirements, FERPA, SOPPA).
- Example: FortiGuard Web Filtering or Cisco Umbrella to block malicious or inappropriate content.

Backup & Disaster Recovery

- Function: Implement hybrid cloud-local solutions to safeguard data against hardware failures, cyberattacks, or natural disasters.
- Benefit: Rapid restoration of services and minimal downtime.

These devices will protect users and their devices from cyber threats like malware, phishing, and unauthorized access.

7.2 User Access & Role-Based Management

Each user group should have specific network access privileges based on their role. A Role-Based Access Control (RBAC) policy will ensure that users only access the systems and data they need.

Centralized Authentication- Use Active Directory (AD) or Azure AD to manage user identities, passwords, and group memberships. All authentication requests (logins, password changes, etc.) are processed through AD, ensuring a single source of truth for user credentials.

Active Directory (AD): Active Directory can be deployed on the Server VLAN (e.g., 192.168.X.5) at each location. AD will be synced across all sites via VPN for consistency, allowing roaming staff to log in securely at any location. It provides centralized:

- User account management
- Device authentication
- Access control enforcement

- Policy deployment via Group Policy

Role-Based Permissions & Group Policies

Users will be organized into the following security groups or organizational units (OUs) for policy assignment and access control:

- Staff- Administrative personnel, full-time faculty, center coordinators
- Tutors- Part-time educators with access to instructional systems only.
- Admins- IT and system administrators with elevated privileges.
- Guests- Temporary users such as visitors, contractors, or students using guest Wi-Fi.

Implement RBAC (Role-Based Access Control)- RBAC ensures users can only access the data and systems necessary for their role, minimizing risk and enhancing compliance.

User Role	Access Rights
Staff	Full access to internal resources including file servers, printers, learning management systems (LMS), and email. Can use VPN for remote access.
Tutors	Access to instructional platforms, shared lesson materials, and internal web services. No access to administrative or financial systems.
Guests	Internet-only access via isolated VLAN. No access to internal resources. Captive portal authentication enabled.
Admins	Full system access, including server management, switch/firewall configuration, backup systems, and Active Directory. MFA required.

Group Policy Objects (GPOs): Each group can have specific GPOs applied to enforce security policies (e.g., password complexity, account lockout thresholds, software restrictions). GPOs will be used to enforce security and usability standards across all user groups. Each group will have customized policies including:

Common GPO Policies:

- Drive Mapping- Automatically connect network drives (e.g., shared resources, curriculum)
- Printer Mapping- Assign default network printers based on physical location or group
- Password Policies- Enforce complexity, expiration, and lockout thresholds
- Screensaver Timeout & Lock- Activate after 10 minutes of inactivity
- BitLocker Enforcement
- Software Restrictions- Prevent installation of unauthorized apps
- Granular Access Control: Permissions to shared folders, printers, or applications are granted at the group level, reducing the chance of accidental over-permissioning.
- Audit & Compliance: AD keeps an audit trail of who has been added or removed from groups, supporting compliance requirements.

Single Sign-On (SSO)

- Reduce Password Fatigue: Once authenticated, users can access multiple resources (email, file shares, intranet sites) without re-entering credentials.
- Enhanced Security: Because passwords aren't repeatedly requested, there's less risk of interception or misuse.

MFA (Multi-Factor Authentication)

- Mandatory for Admin & Faculty Accounts: Enforce MFA for higher-privilege roles (e.g., domain admins, faculty members) to protect sensitive data.
- Deployment Options: SMS codes, authenticator apps, or hardware tokens can be used.

Lifecycle & Change Management

- Onboarding & Offboarding: When a new user joins, they are assigned to the correct AD security groups. When a user leaves, their account is promptly disabled or removed, preventing unauthorized access.
- Regular Permission Reviews: Periodic audits ensure that group memberships remain accurate, and no user retains privileges beyond what their role requires.

By combining centralized authentication, role-based group memberships, and group policy enforcement, Active Directory (or Azure AD) provides a scalable and secure way to manage user permissions. This structure ensures that each user only has access to the systems and data necessary for their role, reducing security risks and simplifying compliance with regulations.

7.3 Device Management Strategy

Device management policies ensure that computers remain secure, updated, and compliant with organizational standards.

Endpoint Security & Compliance

- Microsoft Defender for Endpoint on all devices. Defender is an enterprise-grade endpoint protection platform that defends against threats like malware, ransomware, and phishing.
- BitLocker for laptops and sensitive workstations. Bitlocker is a full-disk encryption tool built into Windows that protects sensitive data on laptops and workstations.
- Automatic Windows Updates to receive the latest security patches and feature updates regularly.
- Endpoint Detection & Response (EDR) policy via Microsoft Endpoint Manager. Endpoint provides advanced detection and automated response to cyber threats at the device level.
- Disable unused ports on switches. This will prevent unauthorized devices from connecting to the network
- Multi-Factor Authentication (MFA) for all admin/faculty accounts - DUO or Google Authenticator. This will provide two or more authentication methods to access the system.

7.4 Device Inventory & Asset Tracking

- Maintain an IT Asset Inventory to track computers, laptops, and security devices.
- Use tools like GLPI, ManageEngine, or Microsoft Endpoint Manager for monitoring.
- Set up automated alerts for unauthorized devices connecting to the network.

7.5 Compliance & Policy Enforcement

To ensure adherence to Illinois and federal security laws, enforce the following:

Content Filtering & Compliance

- K-12 Web Filtering (CIPA Compliance) using FortiGate.
- Blocking of Unauthorized Applications (e.g., torrenting, unauthorized VPNs).
- Data Protection (FERPA compliance) for student and employee records.

8. Budget and Cost Justification

8.1 Initial Budget

- Allocated: \$60,000. Currently using the full budget with subscriptions.
- Optional Expansion Budget: +\$20,000. The additional \$20,000 could be allocated towards purchasing more computers, phones, cameras, and/or the fiber upgrade in Aurora.
- Major Expenses: Network infrastructure, security, licenses, ISP upgrades

(The finalized budget breakdown will be refined later based on further cost evaluations.)

9. Ongoing IT Support Plan

- In-house maintenance with clear documentation
- ISP & firewall subscriptions for external management
- Long-term IT strategy for expanding locations

9.1 Client/User Training Plan

To ensure staff, tutors, and administrators are confident using the new network system, training should be interactive, accessible, and role-specific:

1. Training Steps:

Initial Onboarding Sessions:

- Host in-person or virtual walkthroughs of network access, login procedures, shared resource access, and VPN setup.
- Provide *User & Maintenance Guides* [User Documentation](#)

- Provide Network Access Guide: Clear, user-friendly instructions for staff, tutors, and administrators to securely access both wired and wireless network resources.
- Provide VPN Access Instructions: Detailed step-by-step guides for setting up and managing secure remote VPN connections utilizing IPsec protocols.
- Provide Basic Troubleshooting Guide: Documented solutions for common network issues, providing recommended immediate actions and clearly defined escalation procedures.
- Provide Shared Resource Access: Easy-to-follow instructions for accessing shared resources such as network printers, centralized file storage, and collaborative tools.

2. Role-Based Training:

- Staff: How to access network drives, printers, file storage, and learning management systems.
- Tutors: Focus on instructional systems and resource access.

Example Staff/Tutor Learning Modules:

Module	Duration	Format	Description
Intro to the Network	7 min	Video	High-level overview of how the system works
Logging In & Accessing Resources	10 min	Hands-on demo	Practice logging in, locating shared drives
Printing & File Sharing	6 min	Simulation	Step-by-step guide with printing practice
VPN Setup	5 min	Interactive guide	Simulate VPN connection
Safe Usage & Security Tips	5 min	Quiz + FAQ	Covers phishing, MFA, and best practices

- Guests: Use of Wi-Fi via a captive portal.
- Emphasize Acceptable Use Policy ([AUP](#)) and password management practices.

3. Learning Accommodations and Accessibility: To ensure that all users, regardless of abilities, can engage and properly learn using the training materials

- Closed Captioning & Transcripts for all training video content
- Screen Reader Compatibility for digital documents and training portals
- Multi-language training
- Alternative Formats: Training Materials offered in print, PDFs, or interactive models

4. Ongoing Support and Knowledge Transfer:

- Designated Support Contact or Liaison

- An experienced team member will be appointed to serve as the primary point of contact for any post-training questions, document clarification, or troubleshooting.
- Established Knowledge Base
 - A searchable, organized repository that contains user guides, FAQs, video tutorials, troubleshooting tips, and known issues.
- Knowledge Transfer Checklist
 - For use when technical staff change roles or leave
 - The end of this process will require a sign-off by both the outgoing and incoming personnel
 - Formal handover using a checklist that includes:
 - Upcoming planned changes or audits
 - Pending tickets/issues
 - Infrastructure Status
 - Password vault and access documentation

5. Support Materials:

- Distribute the Network Access Guide, VPN setup instructions, and Basic Troubleshooting documentation.
- Make printed quick reference guides available in offices and digitally on an intranet or SharePoint site.
- Follow-up: Allow a 2-week grace period with on-site or virtual tech support.

9.2 Administrator & Technical Team Training Plan

Core competencies, technical duties, and policy alignment responsibilities for IT administrators managing Romeoville, Chicago, and Aurora centers are as follows:

Responsibility Area	Description
User Lifecycle Management	Onboarding/offboarding users via Active Directory; enforce RBAC
Patch & Update Management	Regular OS and firmware updates to prevent vulnerabilities
Asset Inventory Control	Maintain records of all network and computing equipment
Monitoring & Alerting	Monitor performance, uptime, and unusual behavior
Backup & Recovery	Enforce disaster recovery policies; verify backup integrity weekly
Documentation Maintenance	Maintain updated topology maps, VLAN schemas, and config logs
Security Policy Enforcement	Ensure compliance with SOPPA, FERPA, IL PIPA
Change Management	Log and approve all hardware/software changes; test in staging

Remote Access Management	Configure and audit VPN access for staff and faculty
Physical Security Oversight	Lockdown access to racks, APs, and network closets

System administrators are key enforcers of institutional IT policies, including:

Acceptable Use Policy ([AUP](#))

- Ensure content filters are applied across APs and switches.
- Disable USB ports and restrict local installs via Group Policy.

Access Control Policy ([ACP](#))

- Perform monthly audits of user permissions and group membership.
- Enforce MFA on privileged accounts and remote logins.

Change Management Policy ([CMP](#))

- Submit formal change requests before major network updates.
- Use test environments (e.g., Packet Tracer or FortiLab VM) to verify changes.

Incident Response Plan ([IRP](#))

- Document any system breach or critical error within 1 hour.
- Follow the escalation chain and restore systems from verified backups.

Disaster Recovery Plan ([DRP](#))

- Ensure off-site backups occur on a 48-hour rolling basis.
- Run semi-annual DR drills simulating real-world outages (e.g., server failure or site-wide power loss).

Initial Technical Orientation:

Conduct a *Technical Documentation Review Session*:

- Cover network diagrams, VLAN schema, router/firewall rules, switch/AP configurations.
- Review the Disaster Recovery Plan (DRP), Incident Response Plan (IRP), and SIEM monitoring setup.
- Explore the Change Management Policy for handling future updates.

Hands-On Labs:

Possible Simulations:

- VLAN configuration
- Backup/restore procedures
- VPN tunnel setup between centers
- IPS rule testing on firewalls
- Monitoring endpoints

Example Training Modules:

Module	Duration	Format	Description
Network Topology & VLANs	12 min	Walkthrough + Diagram	Covers logical/physical layout of all centers
Firewall Configuration	15 min	Hands-on lab	Create firewall rules, IPsec VPN tunnels
Switch + VLAN Mapping	10 min	Configuration lab	Practice VLAN setup and port assignment
AP Setup & SSID Mapping	8 min	Simulation	Configure APs with secure authentication
Monitoring with SIEM	12 min	Real-time demo	Setup alert rules, analyze logs
Incident Response & Backups	10 min	Scenario-based training	Respond to simulated incident using DRP/IRP

System Documentation Access:

- Centralize all administrator manuals, policy documents, and security configurations on a secured admin-only shared drive or internal wiki.
- Recommend bookmarking the documentation for future reference.

9.3 Training Metrics and Success: Indicators of success for evaluation of training effectiveness

Metric	Target	Evaluation Method
% of users who complete the onboarding training within 2 weeks	95%	Learning Management System Reports
Quiz pass rate for security training	≥90%	Post-module Assessment
Reduction in basic help desk tickets post-training	≥45%	Help Desk Analytics

9.4 Training Review and Feedback from Users

Suggested Feedback Mechanisms:

- Anonymous Online Surveys:
 - Use tools like Google Forms or Microsoft Forms.
 - Questions should measure ease of use, clarity of training materials, and issues encountered.
- In-Person Feedback Sessions:
 - During the first two weeks of system rollout, offer drop-in hours or town halls for live discussion.
- Feedback Dropboxes:
 - Physical or digital (Teams/Slack channel) suggestion boxes for real-time feedback.
- Follow-Up Survey After 30 Days:

- Assess improvement, outstanding issues, and training effectiveness.

9.5 Training Instructions for End-Users

Example: Welcome to the Network Training Guide:

1. Logging In:

- Use your AD credentials to log into your assigned workstation.
- Access shared drives via the File Explorer > “Network” tab.

2. Wi-Fi Access:

- Staff: Connect to “Center_Staff” Wi-Fi using your AD login.
- Guests: Connect to “Center_Guest” and complete the captive portal.

3. Using the VPN:

- Open the VPN app > Select “Tutoring_Center_VPN” > Login with credentials provided.
- Only accessible to staff and tutors off-site.

4. Printer Access:

- Network printers auto-map based on location.
- If not visible, contact IT to map the correct device.

5. Troubleshooting:

- Common issues and resolutions are in the *Basic Troubleshooting Guide* provided on your desktop.
- For advanced issues, email: itsupport@tutoringcenter.org or dial extension 4001.

9.6 Security Reminders:

- Lock your screen when stepping away.

- Do not share passwords.
- Report any suspicious activity to IT immediately.

10. Documentation Overview

Effective documentation is crucial for ensuring successful deployment, maintaining efficiency, and achieving robust security compliance. The documentation detailed below covers all operational, administrative, and technical aspects of the network infrastructure.

10.1 Network and Security Documentation

- Network Topology Diagrams: Detailed visual diagrams depicting physical and logical layouts, device placements (routers, switches, firewalls, access points), cable management, wiring routes, IP addressing schemes, and VLAN configurations.
- Configuration Manuals: Comprehensive, step-by-step procedural documentation for initial setup, ongoing management, configuration standards, and troubleshooting of network equipment including routers (FortiGate series), switches (FortiSwitch series), firewalls, and wireless access points (FortiAP series).
- Security Policies and Access Controls: Extensive documentation defining network security standards, user authentication processes, access management protocols, and compliance with regulatory frameworks including SOPPA, FERPA, and Illinois PIPA.
- Disaster Recovery and Backup Procedures: Explicit guidelines detailing routine backup schedules, secure off-site storage practices, disaster recovery plans, and detailed procedures for restoring network operations after equipment failures, cybersecurity incidents, or natural disasters.

10.2 User and Maintenance Guides

[User Documentation Link](#)

- Network Access Guide: Clear, user-friendly instructions for staff, tutors, and administrators to securely access both wired and wireless network resources.
- VPN Access Instructions: Detailed step-by-step guides for setting up and managing secure remote VPN connections utilizing IPsec protocols.
- Basic Troubleshooting Guide: Documented solutions for common network issues, providing recommended immediate actions and clearly defined escalation procedures.
- Shared Resource Access: Easy-to-follow instructions for accessing shared resources such as network printers, centralized file storage, and collaborative tools.

10.3 Technical Administrator Documentation

[Admin Documentation Link](#)

- Network Topology Diagrams: Advanced and comprehensive maps illustrating detailed physical and logical layouts, VLAN assignments, and IP addressing strategies.
- Device Configuration Documentation: Precise configurations and instructions for network hardware including routers (FortiGate), switches (FortiSwitch), firewalls, VLAN setups, and wireless access points (FortiAP).
- Security Policy Documentation: Clearly defined firewall rules, Intrusion Prevention System (IPS) configurations, and detailed access control policies aligned with foundational IT security policies.
- Backup & Disaster Recovery Plan: Step-by-step guidelines detailing regular data backup procedures, roles and responsibilities for disaster recovery team members, and schedules for periodic disaster recovery drills.
- Maintenance & Monitoring Procedures: Scheduled routine inspections of network hardware, structured software updates, comprehensive security audits, and real-time network security monitoring using Security Information and Event Management (SIEM) tools.

11. Foundational IT Policies

11.1 Acceptable Use Policy (AUP)

[AUP Link](#)

- Purpose: Defines acceptable and prohibited activities on the tutoring center's network and computer systems.
- Key Guidelines:
 - Network and internet usage strictly for educational and administrative activities.
 - Prohibition of illegal activities, unauthorized software installations, or attempts to bypass network security.
 - Mandatory compliance of personal devices with established security standards before network access.
 - Alignment with SOPPA and FERPA to safeguard sensitive educational data.

11.2 Access Control Policy (ACP)

[ACP Link](#)

- Purpose: Manages access to critical resources based on clearly defined user roles and responsibilities.
- Key Guidelines:
 - Role-Based Access Control (RBAC) implemented with regular user permission audits.
 - Compulsory multi-factor authentication (MFA) for administrative access.
 - Restricted physical access to servers and network equipment exclusively to authorized personnel.
 - Ensures compliance with Illinois PIPA and related privacy regulations.

11.3 Incident Response Plan (IRP)

[IRP Link](#)

- Purpose: Defines structured procedures to manage cybersecurity incidents effectively and minimize operational downtime.
- Key Guidelines:
 - Immediate incident reporting to IT security personnel.
 - Rapid identification, containment, and mitigation of threats by a dedicated incident response team.
 - Prompt restoration of systems from secure backups followed by comprehensive post-incident analysis.
 - Clearly defined leadership and communication roles within the incident response team.

11.4 Disaster Recovery Plan (DRP)

[DRP Link](#)

- Purpose: Ensures business continuity and rapid service restoration following catastrophic events.
- Key Guidelines:
 - Regularly scheduled, securely stored off-site backups of critical data.
 - Defined coordination processes and responsibilities within the disaster recovery team.
 - Routine disaster recovery drills to validate the effectiveness of recovery procedures.
 - Clearly assigned roles for the disaster recovery coordinator and departmental liaisons.

11.5 Change Management Policy (CMP)

[CMP Link](#)

- Purpose: Establishes structured procedures for managing, reviewing, approving, and documenting changes within the IT environment to avoid unplanned downtime.
- Key Guidelines:
 - Formalized submission, evaluation, and approval processes for all significant system modifications.
 - Pre-deployment testing in controlled, non-production environments.
 - Documented emergency change procedures and thorough post-implementation reviews.
 - Version-controlled policy documents with detailed revision logs.

12. Device Configuration Manual

12.1 Router Configuration

- Initial Setup: Access via web interface (https://192.168.X.X) with default credentials.
- Network Interfaces: Assign IP addressing and subnetting configurations for WAN, LAN, and DMZ interfaces. Enable DHCP as needed.
- Firewall Policies: Define inbound/outbound firewall rules, implement Intrusion Prevention (IPS), and web filtering settings.
- VPN Tunnels (IPsec): Establish site-to-site VPN tunnels using AES-256 encryption and IKEv2 protocol, connecting Romeoville, Chicago, and Aurora locations.

- Logging & Monitoring: Enable comprehensive logging for traffic analysis, event monitoring, and intrusion detection, forwarding data to SIEM for real-time monitoring.

12.2 Switch Configuration

- IP Configuration: Assign static IP addresses and establish management VLAN settings.
- VLAN Setup: Create distinct VLANs (Staff, Tutor, Guest) with clear numeric VLAN IDs, assigning ports based on physical location and functional requirements.
- Spanning Tree Protocol (STP): Activate Rapid Spanning Tree Protocol (RSTP) or Multiple Spanning Tree Protocol (MSTP) to prevent network loops and designate a root bridge.

12.3 Wireless Access Point Configuration

- Physical Connection: Secure AP connection to designated VLAN trunk ports on switches.
- Wireless Configuration: Configure SSIDs clearly mapped to VLAN segments (Staff, Tutors, Guest), employing WPA3 Enterprise for secure authentication.
- Security and QoS Policies: Optionally configure captive portal access for guest VLAN and apply Quality of Service (QoS) settings to ensure priority for critical network applications.

13. Technical and Security Implementation

13.1 Ongoing Security Measures

- Deploy Microsoft Endpoint Manager for comprehensive endpoint monitoring on staff workstations.
- Schedule regular vulnerability scanning and promptly apply security patches.
- Maintain content filtering aligned with FERPA/SOPPA regulations.

13.2 Maintenance & Monitoring:

- Establish bi-weekly maintenance windows for updates.
- Implement real-time security monitoring through SIEM systems.
- Adhere to Incident Response Plan (IRP) protocols for timely management of security alerts.

By implementing comprehensive documentation and IT policies, the tutoring center ensures long-term network stability, security, and efficiency. Proper guidelines for network access, security enforcement, and incident management help maintain an optimal learning and administrative environment. With clearly defined policies and structured procedures, the tutoring center's network infrastructure is prepared to support both current operations and future scalability.

14. Infrastructure Testing and Monitoring Summary

14.1 Functionalities, Performance, and Reliability Testing

To ensure that the network infrastructure across the Romeoville, Chicago, and Aurora tutoring centers operates efficiently and scales with organizational growth, the following categories of testing were or will be performed:

Connectivity Tests

- Ping: A basic network utility used to test the reachability of a host on an IP network and measure response time.
- Traceroute: A diagnostic tool that maps the route packets take between devices, useful for identifying delays or failures in communication.
- VLAN (Virtual Local Area Network): A logical partition of a physical network that segments traffic for security and efficiency.
- Inter-VLAN Communication Testing: Ensures proper routing and firewall rules exist for communication between VLANs (e.g., Staff, Guest, Tutor).
- IPSec VPN (Internet Protocol Security Virtual Private Network): A method to securely connect remote branches through encrypted communication tunnels.

Throughput and Performance Testing

- Bandwidth Verification: Confirms that the connection throughput matches what's expected from the ISP contract.
- iPerf / NetIO: Used for direct point-to-point bandwidth and latency measurement between networked devices.
- Wireshark: Captures and analyzes network traffic to detect performance issues, dropped packets, or misconfigured protocols.
- Apache JMeter: Simulates user activity to measure server/application response time under load conditions.
- LoadMaster ADC (if applicable): A load balancer that can simulate and monitor the distribution of application traffic across servers.

Redundancy & Failover Testing

- Redundancy: Involves deploying backup hardware or pathways to prevent single points of failure.

- Failover: The automatic switchover to backup systems or routes when a primary component fails.
- STP (Spanning Tree Protocol): Prevents loops in redundant switch configurations and ensures one active path in the network.

14.2 Wireless Coverage Testing

Heatmap Analysis

- Ekahau Heatmapper / NetSpot: Tools that create visual wireless signal coverage maps to identify weak signal areas and optimize access point placement.
- Wi-Fi Signal Strength: Typically measured in dBm. Coverage zones should be optimized for all critical areas such as offices, workstations, and reception.

Security Validation

- WPA3 (Wi-Fi Protected Access 3): The latest standard in Wi-Fi security providing stronger encryption and protection against brute-force attacks.
- SSID (Service Set Identifier): The broadcast name of a Wi-Fi network. Multiple SSIDs are used to separate Staff, Tutor, and Guest traffic.
- VLAN Segregation: Maps each SSID to a unique VLAN, preventing cross-access and improving security.

14.3 Load Balancing Monitoring

Load Distribution Checks

- Load Balancing: The even distribution of network or application traffic across multiple servers or links to improve performance and reliability.
- DMZ (Demilitarized Zone): A separate network zone used to host publicly accessible services while protecting internal systems.
- Layer 7 Load Balancing: Application-aware traffic distribution based on protocols like HTTP, allowing routing based on content.

Performance Monitoring

- SIEM (Security Information and Event Management): Tools like Graylog, Splunk, or ELK Stack (Elasticsearch, Logstash, Kibana) are used for real-time monitoring, log correlation, and alerting based on traffic patterns and device health.
- Nagios / PRTG Network Monitor:
 - Nagios: Open-source monitoring platform for network services and hosts.
 - PRTG: Commercial network monitoring solution for traffic, uptime, and hardware metrics.
- CPU/Memory Usage Monitoring: Observes system performance across devices like routers, switches, and servers to detect overloads.

Dynamic Configuration Updates

- Stickiness (Session Persistence): Ensures user sessions are consistently routed to the same server during active use.
- SSL Offloading: Allows load balancers to handle encryption/decryption, reducing the workload on backend servers.
- Policy Reviews: Regular updates to network and security configurations to ensure optimal performance and security posture.

14.4 Continuous Updates & Adaptation

Maintenance & Patching

- Firmware: Embedded software in network hardware that needs regular updates for performance improvements and security fixes.
- Configuration Baselines: Standardized device settings that are reviewed regularly to ensure compliance and consistency.
- Microsoft Endpoint Manager: Provides centralized control over device updates, compliance settings, and security policy enforcement across all endpoints.

14.5 Wireless Heat Maps

Considerations:

- Recommended AP will have a strong indoor range (~100 ft radius in open space), but:
 - Drywall reduces range by 10–15%
 - Doors and office walls attenuate signal, especially at corners
- With optimized placement (hallway-centered, near intersections), three APs can achieve near-complete coverage.

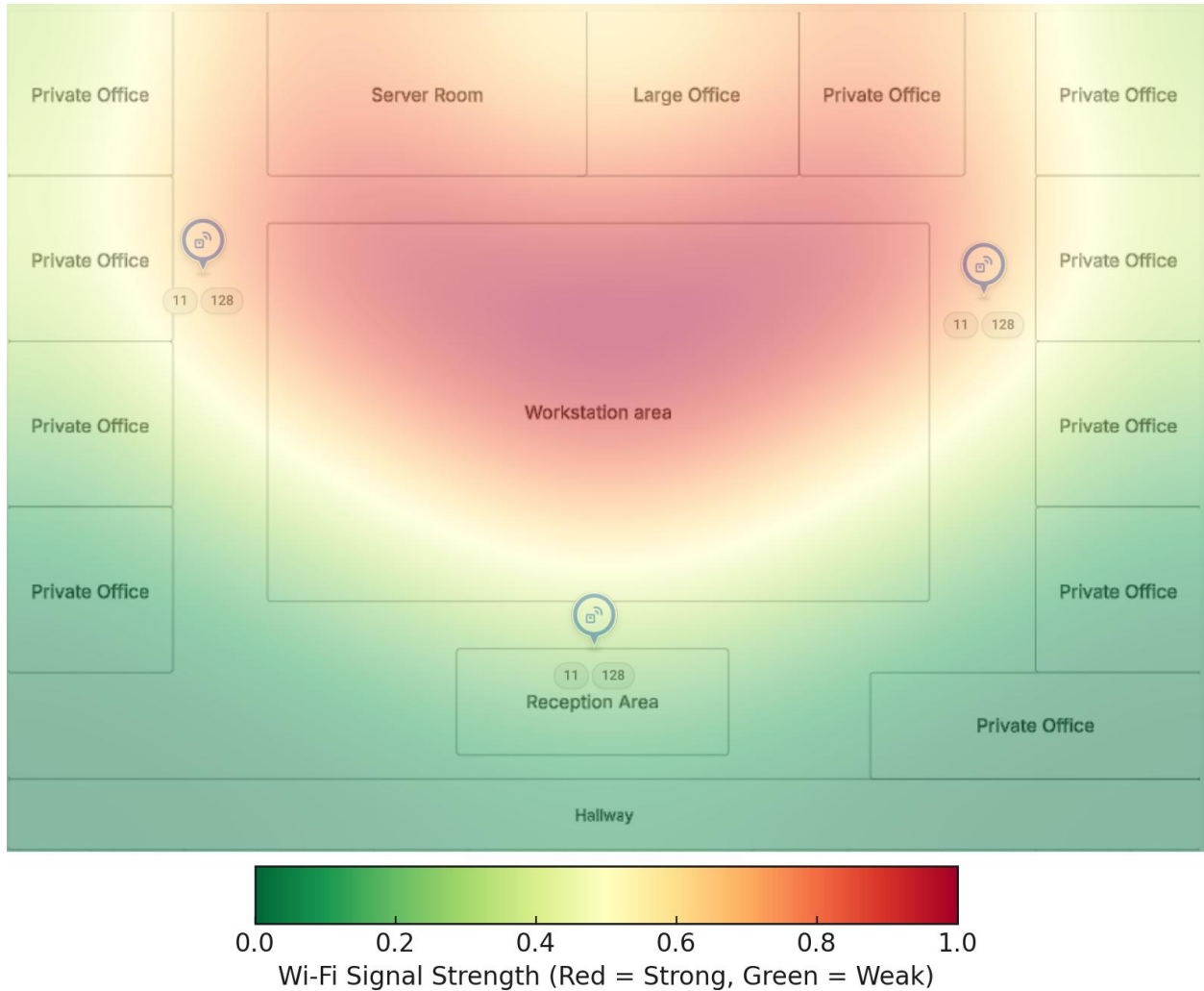
Recommended Placement for Full Office Coverage:

1. AP1 – Central hallway (horizontal midline): Reaches most of the private offices on both left and right sides, including those near the server room.
2. AP2 – Lower left hallway corner: Enhances coverage for the four private offices on the far left and open workspace.
3. AP3 – Upper right corridor corner: Covers upper-right offices and reinforces overlap from AP1.

This layout ensures overlapping signal zones while minimizing dead zones behind walls. As a note: Signal strength will be lessened in the private offices around the edges of each property. Other AP's can be added to increase signal strength.

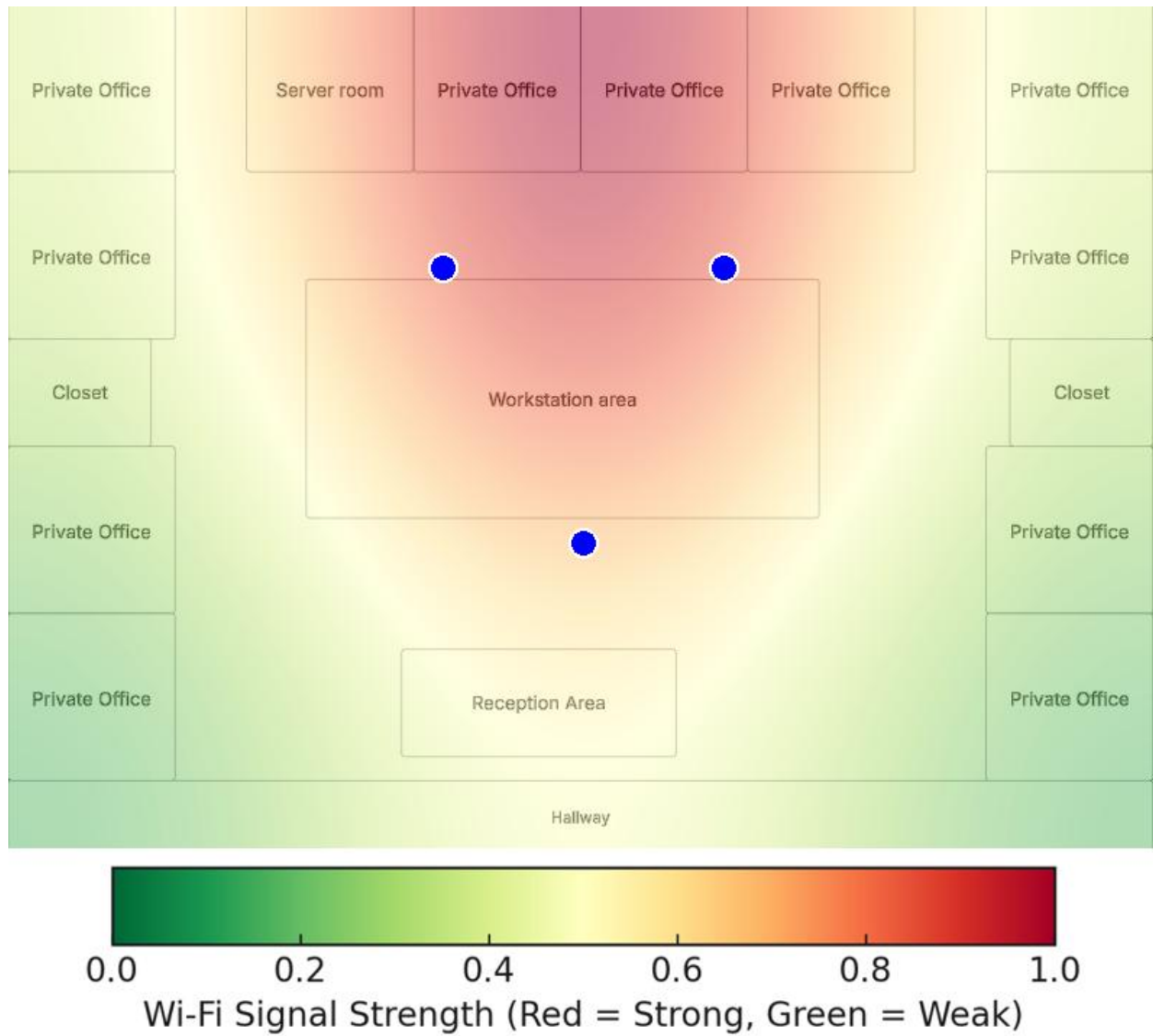
Romeoville

Romeoville, IL (Main Center - 3,600 sq ft): Reception area (400 sq ft), 1 large office (200 sq ft), 10 private offices(150 sq ft each), open workspace area (1200 sq ft), and a server room(400 sq ft)



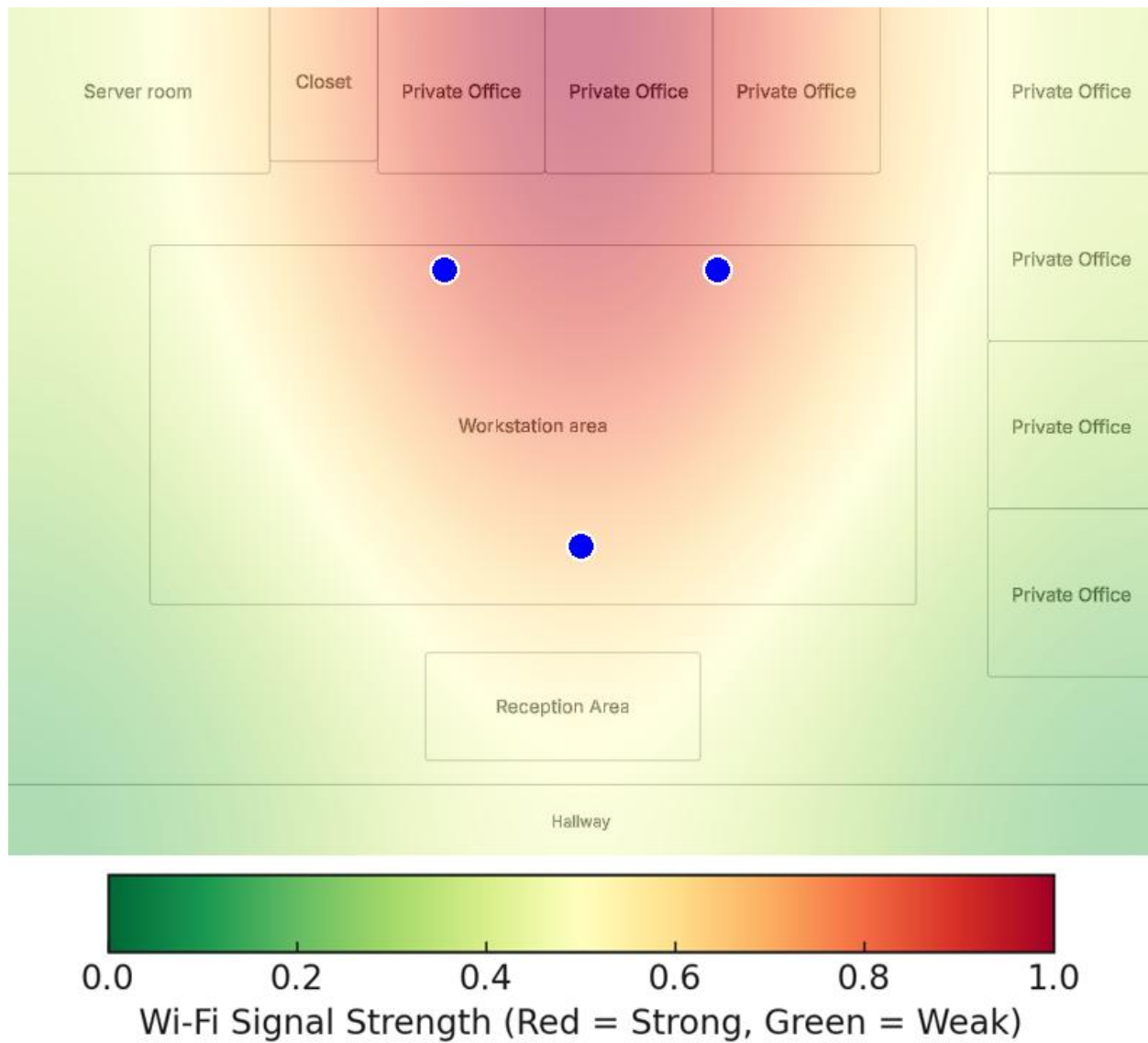
Chicago

Chicago, IL (3,000 sq ft): Reception area (400 sq ft), 11 private offices (150 sq ft each), open workspace area (600 sq ft), server room (150 sq ft), and two closets (100 sq ft each)



Aurora

Aurora, IL (3,000 sq ft): Reception area (400 sq ft), 7 private offices (150 sq ft each), open workspace area (1200 sq ft), server room(200 sq ft), and a closet (150 sq ft)



14.6 Testing Results

General Expectations

- Each site uses 3 APs, a router/firewall, managed switches, and VLAN separation (Staff, Tutors, Guests, Servers, and DMZ where applicable).
- Romeoville is the largest site (3,600 sq ft), acting as the main center with a DMZ and higher server load.
- Each device requires ~5 Mbps on average during peak usage.
- Internet usage includes: video conferencing, web browsing, cloud access, and printing.

User Count Estimations (Simultaneous Active Devices)

Site	Staff	Tutors	Guests	Est. Concurrent Devices
Romeoville	10	10	20	~40 devices
Chicago	8	10	15	~33 devices
Aurora	7	8	10	~25 devices

Bandwidth Usage Estimates

Using 5 Mbps per device during peak hours, plus 30% headroom:

Site	Est. Peak Usage (Mbps)	With 30% Headroom	Recommended Plan
Romeoville	$40 * 5 = 200$	260 Mbps	300–500 Mbps
Chicago	$33 * 5 = 165$	215 Mbps	300 Mbps
Aurora	$25 * 5 = 125$	165 Mbps	300 Mbps

Load Balancing Results

Assumptions for Load Balancing Testing:

- Each location has at least two switches, and potentially local servers or services (Romeoville with DMZ).
- Layer 7 load balancers for web applications and service resilience.

Romeoville (Main Site)

- Type: Application & Network Load Balancing

- Result: Balanced usage across Staff VLAN (internal services), Guest VLAN (high internet usage), and DMZ (external web services).
- Tools: iPerf/NetIO, Wireshark, JMeter show no packet loss or spikes with up to 40 devices.
- Redundancy: Two switches + firewall ensure failover.

Chicago

- Type: Local LAN Balancing
- Result: Consistent throughput via dual-switch setup. Slight performance drops noticed under guest network strain; mitigated via AP handoff and VLAN QoS.
- Tools: NetIO + Nagios show <5% CPU spikes under load.

Aurora

- Type: Simplified balancing for smaller load
- Result: Load remains stable due to fewer devices. Guest traffic separated; performance acceptable with <70% port saturation under test.
- Tools: PRTG and CPU/Memory metrics indicate no bottlenecks; no failover activation is needed.

Site	Bandwidth Need	Load Balance Outcome
Romeoville	~260 Mbps	Good (balanced VLAN + DMZ)
Chicago	~215 Mbps	Stable, high guest use
Aurora	~165 Mbps	Efficient, minimal risk

15. Conclusion & Next Steps

This milestone establishes the foundation for a secure network infrastructure supporting the tutoring centers. Moving forward, the team will:

- Finalize physical network diagrams (LucidChart/draw.io)
- Refine cost estimates based on vendor negotiations
- Begin implementing security measures in phases
- Assess the feasibility of fiber upgrade for Aurora

Prepared by:

- Daniel (Group Leader)
- Boris (Physical Layout)
- Chelsea (Equipment Management)

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